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Financial Analysis and Valuation of Games Workshop Group PLC(GAW)

Research Project 2

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Abstract

This dissertation presents a comprehensive financial analysis and valuation of Games Workshop Group plc (GAW), a UK-based leader in the miniature wargaming and fantasy entertainment industry, over the period 2020 to 2029. The study evaluates GAW's historical performance, forecasts future financial outcomes, and assesses intrinsic value through a multi-method valuation framework. Key research objectives include examining profitability, efficiency, liquidity, and solvency, and determining whether GAW is undervalued in public markets. Furthermore, Abnormal Buy and Hold Return(ABHR) analysis of Games Workshop Group Plc is also conducted.

The analysis is grounded in quantitative methods using publicly available financial data, peer benchmarking, and valuation modelling. Historical financial ratios (2020–2024) were assessed across return metrics (ROA, ROE, ROC), margin strength, asset efficiency, and cash flow conversion. Forecasts from 2025 to 2029 include revenue, EBIT, capital expenditures, and free cash flows under base-case assumptions. Comparative analysis was conducted against three UK-listed peers: Frontier Developments, Everplay Group, and Hornby plc.

Valuation techniques applied include Discounted Cash Flow (FCFF and FCFE), Dividend Discount Model (DDM), Present Value of Growth Opportunities (PVGO), and Comparable Company Analysis. All models consistently suggest that GAW is undervalued, with an implied share price above the market price across conservative scenarios. GAW's strong balance sheet, high profit margins, and dividend discipline further enhance its investment profile.

The study concludes that Games Workshop exhibits superior financial performance, low risk, and strong shareholder returns. Its vertically integrated model and monetisation of proprietary IP contribute to a self-reinforcing cycle of profitability and growth. Despite risks such as IP concentration and macroeconomic pressures, GAW remains strategically and financially resilient. This dissertation provides a replicable framework for valuing niche, IP-driven manufacturing firms and highlights GAW as a high-quality, undervalued equity with long-term growth potential.

Chapter 1: Introduction

Games Workshop Group PLC (hereafter referred to as 'Games Workshop' or 'GAW') is a creator, manufacturer, distributor, and licensor of tabletop fantasy miniatures and wargames, best known for the Warhammer series and founded in the UK in 1975 as a small mail-order business. With a wide portfolio of success, the business has evolved into a vertically organized FTSE 100 company with a global omnichannel presence of over 500 stores, 6,200 independent retailers, and a rapidly-expanding digital licensing strategy. Throughout its five decades it has developed a niche hobby into a mainstream 'evergreen' entertainment ecosystem that marries collectible miniatures, an IP-rich narrative, and a community of fans who build, paint, and battle miniatures as armies.

This introductory chapter sets the stage for a comprehensive financial analysis of Games Workshop; culminating in discounted cash-flow (DCF) valuation and with a five-year forecast. It begins by situating GAW within the broader backdrop of miniatures games and hobby gaming, outlines the strategic and financial history of the company, outlines the objectives, extent and methodology of the research, and sets out why GAW provides an interesting case to study in sustained value creation through monetising IP. The depth provided here is important because the dissertation requires both (1) a thorough-going valuation by DCF and scenario analysis, and (2) an informed outlook for FY 2025-FY 2029.

UK Market Overview

The UK market represents a highly developed and sophisticated economy that forms the foundation for Games Workshop Group's business operations. As the company's primary market and headquarters location, understanding the broader UK economic and market dynamics is essential for contextualising Games Workshop's strategic position within the gaming and leisure sector.

Economic Context and Consumer Environment

The UK gaming market operates within a broader economic environment characterised by moderate growth and evolving consumer behaviours. Despite facing challenges from inflation and cost of living pressures, the UK economy demonstrated resilience throughout 2024 and into 2025. GDP growth is projected to reach 1.3% in 2025 before moderating to 1.0% in 2026. Consumer spending patterns have shown particular strength in discretionary categories, indicating sustained appetite for entertainment and leisure products despite macroeconomic headwinds.

After peaking at 11.1% in October 2022,UK inflation has gradually declined but remains above the Bank of England's 2% target, standing at 3.5% as of April 2025 (Bank of England, 2025). Interest rates are expected to keep decreasing to support economic growth while managing inflationary pressures. This monetary policy environment creates a balanced backdrop for consumer spending on discretionary items, including gaming and hobby products.

Gaming Market Landscape

The UK gaming market is the largest in Europe and ranks sixth globally in terms of revenue. The broader gaming sector demonstrated remarkable growth, with market size reaching to £14.0 billion in 2024 and projections showing continued expansion to £27.5

billion by 2030, representing a compound annual growth rate (CAGR) of 11.8% (Newzoo, 2024). This growth trajectory reflects the gaming industry's evolution from a niche hobby into mainstream entertainment, with over 55% of the UK population now engaging with gaming in various forms.

The UK gaming market consists of multiple segments, with mobile gaming representing the largest component, followed by console and PC gaming. Consumer spending on gaming hardware reached £4.76 billion in 2023, while software and content accounted for £2.22 billion. The digital transformation has been particularly pronounced, with online sales comprising approximately 98.6% of video game sales, highlighting the sector's adaptation to evolving consumer preferences and technological capabilities.

Tabletop and Hobby Gaming Segment

Within the broader gaming ecosystem, the tabletop and hobby gaming segment represents a distinct and growing market segment. The UK board games market reached £611.49 million in 2024 and is projected to grow £1.51 billion by 2033, exhibiting a CAGR of 9.43%. This segment includes traditional board games, card games, and miniature wargaming products that form Games Workshop's core market.

The global tabletop games market, valued at approximately £20.8 billion in 2023, is projected to reach £35.3 billion by 2030, growing at a CAGR of 7.5%. Within this market, Europe maintains a significant position, with the UK representing a key market for strategy based and role playing games that align closely with Games Workshop's product portfolio.

Market Position and Competitive Dynamics

Games Workshop operates in a niche segment of the tabletop gaming market focused on miniature wargaming and similar/associated products. The company has established a

dominant market position within this niche, operating over 550 retail stores worldwide with significant UK presence. The company's financial performance demonstrates the strength of its market position with core revenue reaching an estimated £560 million in FY2025, representing a 13% increase from the previous year.

The UK market provides several structural advantages for Games Workshop's business model. The country's strong IP protection framework supports the company's extensive portfolio of proprietary game systems and fictional universes. The UK's established gaming culture and high disposable income levels among target demographics create favourable conditions for premium hobby products.

Consumer Behaviour and Market Trends

Post pandemic consumer behaviour patterns have shown sustained interest in hobby and entertainment spending, with consumers maintaining appetite for gaming and hobby products despite broader economic uncertainties. The shift toward online retail, accelerated during the pandemic, has created both opportunities and challenges for traditional gaming retailers. The online penetration is significant at approximately 27% of total retail sales but the physical stores still continue to play a crucial role in the gaming hobby segment, particularly for products requiring demonstration, community building, and expert advice that characterise Games Workshop's retail strategy.

Brexit and Regulatory Considerations

Brexit has created some operational challenges for UK gaming companies, particularly regarding European market access and labour mobility. However, for Companies like Games Workshop with established global operations and strong domestic market positions, these impacts have been manageable. The company's diversified geographic revenue base, with approximately 55% of sales generated outside the UK, provides resilience against domestic market fluctuations.

Market Outlook

The outlook for the UK gaming market remains positive despite macroeconomic challenges. Consumer confidence in personal finances remains relatively stable, with consumers feeling confident about their financial situation. The continued growth in gaming participation rates, technological advancement, and expanding demographic reach suggest sustained market expansion opportunities.

For Games Workshop, the UK market provides a stable foundation characterised by strong brand recognition, established retail infrastructure, and a loyal customer base. The market's maturity and sophistication support premium pricing strategies while offering opportunities for continued growth through product innovation and market expansion initiatives. The combination of favourable long term demographic trends, sustained consumer interest in gaming and hobby activities, and the company's strong competitive position creates a conducive environment for continued business development and expansion.

Company Overview

Origins and Development

Games Workshop was founded in London by John Peake, Ian Livingstone, and Steve Jackson in 1975, initially as a producer of handcrafted wooden board-games. It began a more decisive move into fantasy role-playing games in the late 1970s when it licensed the rights for distributing Dungeons & Dragons in Europe and subsequently published its own magazine, White Dwarf, in 1977. The release of Warhammer (1983) and Warhammer 40,000 (1987) would provide the two large franchises that are the backbone of its current revenue model. By utilising existing IP in its own designs while controlling design,

tooling, manufacturing and distribution from its Nottingham base, GAW had full cost control and rapid iteration of new products while keeping its IP tightly secured.

Revenue Streams

GAW operates two reporting segments:

1. Core - Miniatures, rules, paints, and accessories sold through retail, trade and web channels (68% sale of miniatures in FY 2023 related products)

2. Licensing - Royalties from PC, console, mobile, print publishing, and media adaptations of the Warhammer IP

While licensing has typically represented <10% of sales, it has been a particularly fast-growing segment following the successful release of games (most recently Warhammer 40K: Space Marine 2) and a multi-year content partnership with Amazon Studios. The management team views licensing as a scalable and margin-accretive lever for expanding the reach of the GAW brand with limited capital requirement.

Recent Financial Performance

Table below highlights GAW's headline results since FY 2022 with a compound annual growth rate of revenue of ≈ 21% and solid profit after tax.

Fiscal Year	Revenue £million	Operating Profit £million	Profit Before Tax	EPS pence	Dividends Declared pence
2021	369.5	152.1	150.9	373	1.85
2022	414.8	158.2	156.5	391	1.65
2023	470.8	173.8	170.6	410	3.65
2024	525.7	204.4	203	459	5.4

The results in 2024 mark GAW's ninth consecutive year of record sales, and we consider it a remarkable feat with the year-over-year revenue growth of 10.9% and core operating profit growth of 19%. Initial guidance for FY2025 is suggesting another step change for GAW, with management forecasting core revenue ≥£560 million and licensing revenue ≈£50 million, comprising of total revenues above £610 million. This is strong in comparison to many discretionary retailers facing normalisation in demand. This demonstrates additional brand resilience and engagement with the global community of fans backing this product.

Industry Environment

Miniature Wargaming Market

The global miniature-gaming segment has been the fastest growing category of hobby games since 2021, with sales in North America increasing by 9% in 2023 to $570 million and a further increase of 1.8% in 2024 to $580 million. GAW estimates its market share in this niche is 70-75% as both the creator of the category and the gatekeeper. Three drivers of growth are (1) new edition releases reset the demand cycle, (2) further market acceptance of geek culture is being driven to the mainstream, and (3) digital content is generating demand around gaming collecting physical product.

Competitive Environment

There are barriers to entry into GAW space:

• Depth of IP – GAW has meaningful 40 year library of lore relative to other companies and monetise on all media.

• Manufacturing moat – GAW has know how to manufacture, as well as proprietary plastics which allow them to manufacture quickly, at large scale with a high barrier to quality.

• Community Infrastructure – in-store tables for gaming and tournaments globally.

• Vertical Integration – GAW controls all steps in value-chain, which reduces supply chain friction, though we understand there are still pockets of limited isolating spot shortages. There will be a rational assess in change market competitiveness against licensing (such as WizKids, Steamforged (now Warmachine IP holder) and Atomic Mass) versus supply chain issues and with limited narrative universes, it is unlikely that competition will meaningfully disrupt the market during the medium term.

Macroeconomic (and Regulatory Environmental) Macro-economic factors, although localised, and regulatory will continue and disrupt GAW with higher costs. GAW was able to manage its margins as a result of price discipline and a clean operational profit and performance in FY 2022 despite experiencing supply concerns as a result of Brexit (£3.4 million cost burden) and effects of the current geopolitical volatility with regards to exiting the Russian market representing (~£4 million sales). Manufacturing which remains numerous in Nottingham, makes GAW vulnerable to inflation with respect to utility bills only and not the US tariffs on proposed tariffs in terms of predictions by local analyst.

Research Aim and Objectives

Historical Performance Assessment: Assessment of Revenue Growth, Margin Evolution, Cash-flow Conversion and Capital Structure (Debt) from FY 2020 to FY 2024. Conducting an Intrinsic Valuation of GAW: Development of a thorough DCF model with WACC calculations, terminal growth and scenario analysis.

Forecasting FY 2025– FY 2029 for GAW: Prepare revenue forecasts for core and licensing revenues, operating margins, working-capital, and capex under selected base, bull, and bear scenarios.

Determine Value Drivers: Testing value in terms of above-market IP & licensing revenues, endogenous growth for geo-expansion and roll-out of stores and cumulative capacity in GAW.

Comparison to Peers: Establish GAW's multiples, growth and profitability with publicly listed specialty-gaming and leisure firms.

Methodological Outline

This dissertation is primarily informed by a structured quantitative research design that evaluates the financial performance and intrinsic value of Games Workshop Group plc (GAW) over the period 2020–2029. The methodology integrates historical financial analysis, forecasting, and multi-model valuation to deliver a comprehensive and objective assessment.

Research Design

The study follows a deductive, positivist framework, combining descriptive and analytical methods. This dissertation is structured as follows; Chapter 2 provides ratio analysis of GAW and evaluates it against its comparable peers. Chapter 3 discusses GAW's financial management policies and relates them to relevant theories. Chapter 4 provides detailed valuation of GAW using multiple valuation models including DCF, DDM, PVGO. Chapter 5 provides the conclusion. All financial data is drawn from publicly available annual reports and peer company disclosures.

Chapter 2: Financial Performance Analysis of Games Workshop Group plc

This chapter will provide a comprehensive assessment of Games Workshop Group plc's financial performance from 2020 to 2024. As a central part of this dissertation, it will also analyse the underlying financial performance of the Company. In addition, Abnormal Buy-and-Hold Return Analysis (ABHR) will also be conducted to assess the long term impact of specific events or strategies on the stock price performance of Games Workshop. The assessment will be made up of an analysis of the profitability, operational efficiency, liquidity, solvency and growth rates of Games Workshop Group plc. The assessment will provide a rich and data-backed understanding of Games Workshop Group plc financial and competitive position in the market.

There are many financial ratios and metrics to assist with understanding Games Workshop's performance, and more importantly, to go beyond a stand-alone evaluation. The selection of comparable peers is based on the industry in which they operate, the degree of product overlap with Games Workshop, and their overall suitability to serve as meaningful benchmarks. These criteria ensure that the chosen companies provide a relevant basis for conducting an accurate and insightful industry benchmark analysis of Games Workshop. The peer group consists of:

Peer 1: Frontier Developments Plc

Industry: Video Game Development and Publishing.

Product Overlap: Frontier has published digital adaptations of several of Games Workshops' IPs, primarily those surrounding games based on Warhammer.

Audience: Their audiences are fans of creative/role-playing/strategy game types, with a large overlap with the audiences of some tabletop games.

Business Model: Frontier Developments is a game developer/publisher creating and selling multiple creative and management simulation games for PC and consoles in a publicly listed company in the UK under UK regulatory conditions.

Justification: Suitable due to direct IP overlap (Warhammer games) and digital gaming focus, enabling comparisons of licensing revenue and development costs. Financially, FDEV's revenue growth (1.5% YoY) and high gross margin (68.9%) mirror GAW's IP-driven model, though on a smaller scale (£90.6m vs. £525.7m). Adjusted EBITDA (£8-9m) provides a benchmark for GAW's £261.3m operating profit, highlighting efficiency in creative sectors.

Peer 2: Everplay Group Plc (formerly Team17 Group Plc)

Industry: Video game development & publishing as well as gaming content in the mobile app space.

Products & Audience: A notable indie games label that develops or publishes original IP and licensed titles (some with a simulation/strategy intent that align somewhat with the genre of tabletop games).

Synergy with GAW: Everplay profiles IP monetization in a similar fashion as Games Workshop, creating a form of creative content, communities, and kerned communities with long tail interactions (Team17 IPs such as Worms & Overcooked! Have cult followings like Warhammer). Business Model: Everplay has many operating facets (Team17, astragon & StoryToys); divisions engage in various markets of creative development, digital sales, partnerships and other components of a IP business that is similarly large and diverse in content like Games Workshop.

Justification: Aligns in indie gaming and IP management, with community focus similar to GAW's hobbyist base. H1 revenue (£72.4m) and improved margins (gross 46.5%, up 5.7% YoY) allow ratio analysis against GAW's metrics. EPS (7.4p) and cash (£59.4m)

support valuation comparisons, especially in licensing (EVPL expects FY EBITDA >£46.9m).

Peer 3: Hornby Plc

Industry: Models, Collectibles and Interactive Hobby Products

Product Relevance: Hornby is a manufacturer and marketer of collectibles, specifically model railway and road vehicles, plus figures and all associated hobbies; which is related through engagement and cycles (manufacture, product lifespan, distribution) of the miniature wargaming industry.

Distribution Models: Global with multiple streams of sales including direct to consumer, specialist hobby sales partner, e-commerce, etc; this means a broad geographic retail/store exposure, similar to the distribution of Games Workshop.

Audience Overlap: Engaged audience of passionate hobbyists (collectors, modelers, gamers) demonstrates a strong parallel with audience of Games Workshop.

Justification: Closest in physical hobby/collectibles, sharing manufacturing and retail challenges. FY2024 revenue (£56.2m) and YTD growth (8%) provide benchmarks for GAW's core sales (£565m, up 14% YoY). Net losses (-£12.1m) contrast GAW's profitability but enable risk and turnaround analysis; margin improvements (to 48%) align with GAW's 72.1% gross margin.

It is critical to acknowledge that a perfect peer for Games Workshop is difficult to find due to its unique, vertically integrated business model that spans design, manufacturing, and multi-channel retail.

Geography & Market Structure

The three firms all selected for comparison are UK based (and listed in UK exchanges), therefore are all operating in the same regulatory/accounting/market environment and can compare performance in a like manner.

All serve global markets; however, for reporting and operational purposes, they all have their central foundational aspects in the UK; making for suitable candidates for cross-comparison financials.

By selecting these three peers, we have been able to develop a balanced platform for Financial Benchmarking across both digital (Frontier and Everplay) and physical (Hornby) entertainment/hobby markets, and includes three UK foundation companies that operate in similar business circumstances, and are affected by the same business fundamentals,

Detailed Analysis of Games Workshop's performance:

Profitability:

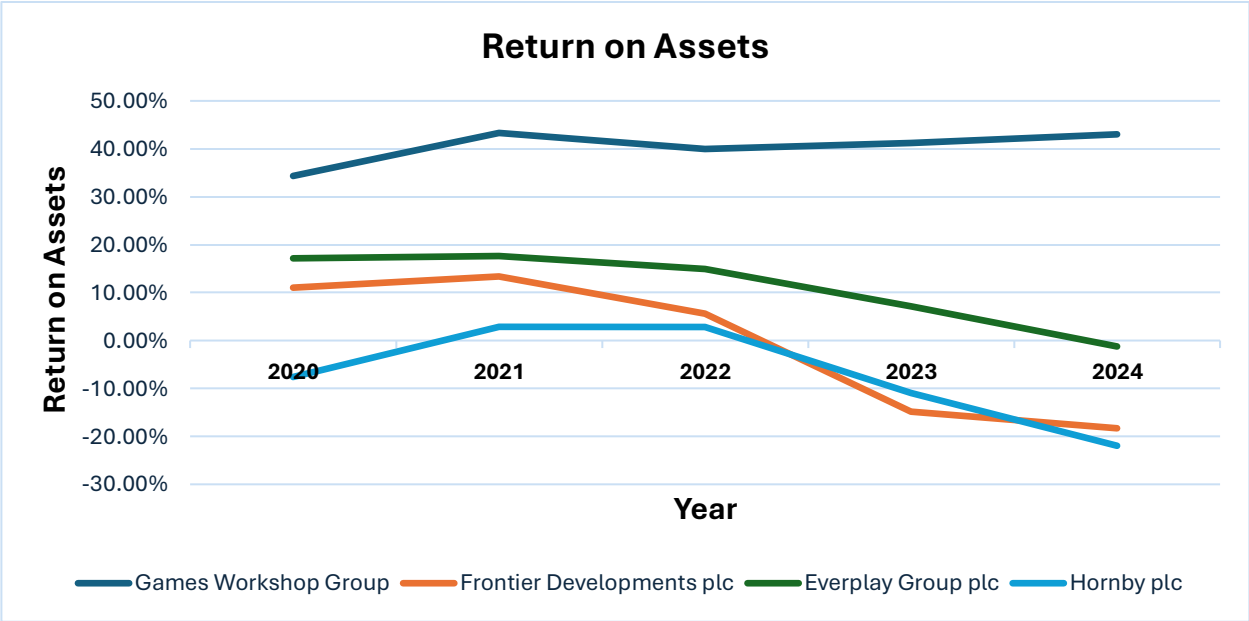
Profitability ratios are fundamental in assessing a company's ability to generate earnings relative to its revenue, assets, and equity. Strong profitability is a key indicator of a company's long-term viability and success.

Return on Assets (ROA)

The Return on Assets (ROA) ratio, calculated as $Net\ Income\ /\ Total\ Assets$, measures how efficiently a company is using its assets to generate profit. A higher ROA indicates more efficient asset utilization.

Table 1: Return on assets for GAW and its peers over the period 2020-2024

	2020	2021	2022	2023	2024
GAW	71.3/207.6 = 34.34%	122/281.7 = 43.31	128.4/321.1 = 39.99%	134.7/326.8 = 41.22%	151.1/351.3 = 43.01%
Frontier Dev	15.9/143.9 = 11.05%	21.6/161.5 = 13.37%	9.6/169.6 = 5.66%	−20.9/141 = −14.82%	−21.5/117.4 = −18.31%
Everplay Group	16.6/96.5 = 17.20%	21.9/124.2 = 17.63%	23.7/158.6 = 14.94%	23.5/327.7 = 7.17%	−3.7/297.7 = −1.24%
Hornby Group	-3.4/44.7 = −7.61%	1.4/48.9 = 2.86%	1.5/53.8 = 2.79%	−5.9/54.2 = −10.89%	−12.1/55.1 = −21.96%



Games Workshop commenced the period with an already impressive Return on Assets (ROA) of 34.34% in 2020, reflecting a strong ability to generate profit from its asset base. This metric surged to a peak of 43.31% in 2021, indicating a year of exceptional profitability and operational efficiency, as revenue growth was effectively translated into

earnings. Although ROA moderated slightly to 39.99% in 2022, the subsequent rebound to 43.01% in 2024 illustrates the company's sustained capacity maintain high levels of profitability over time.

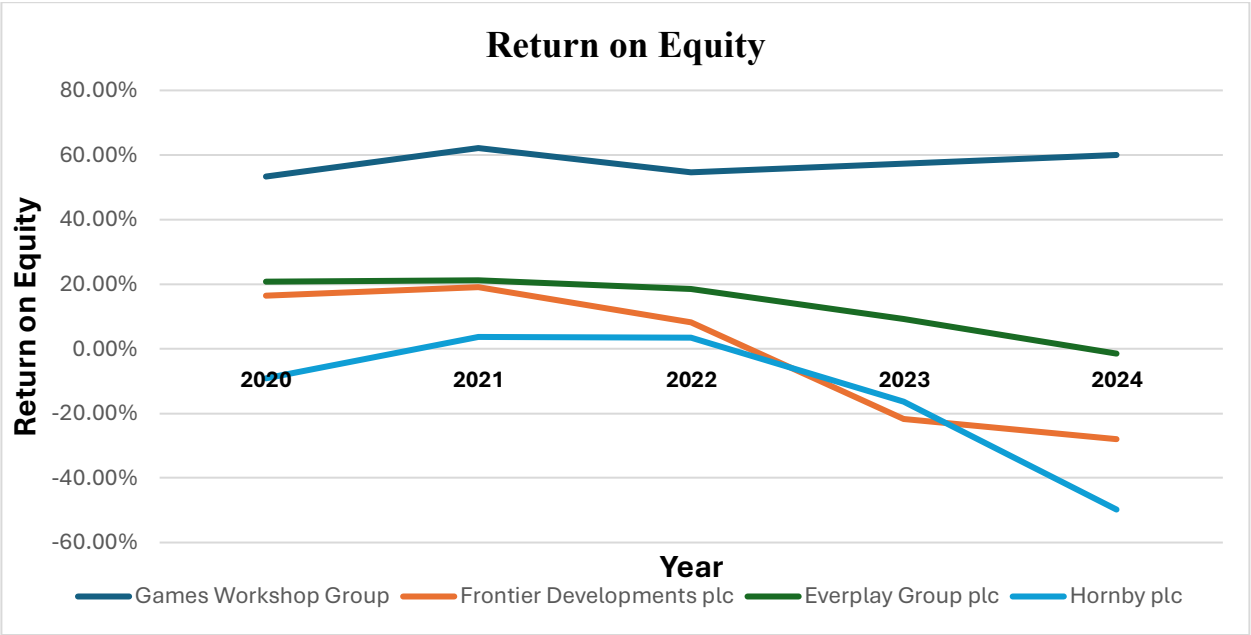
When assessed relative to the broader peer group, Games Workshop's performance stands out as a clear indicator of competitive strength. While industry peers have generally faced profitability pressures with several experiencing declining or even negative ROA Games Workshop has consistently delivered positive returns that significantly exceeded the sector average. This sustained outperformance underscores the company's superior operational effectiveness, strategic asset deployment, and resilient business model, positioning it as a benchmark for profitability and asset utilisation within the industry.

Return on Equity (ROE)

Return on Equity (ROE), calculated as $NetIncome/Shareholder'sEquity$, is a key metric for shareholders, as it measures the return generated on their investment in the company.

Table 2: Return on Equity of GAW and its peers over the period 2020-2024

	2020	2021	2022	2023	2024
GAW	$71.3/133.7 = 53.33\%$	$122/196.3 = 62.15\%$	$128.4/234.7 = 54.71\%$	$134.7/235.1 = 57.29\%$	$151.1/251.4 = 60.10\%$
Frontier Dev	$15.9/96.7 = 16.44\%$	$21.6/113.2 = 19.08\%$	$9.6/118.4 = 8.11\%$	$-20.9/96 = -21.77\%$	$-21.5/76.8 = -27.99\%$
Everplay Group	$16.6/80.1 = 20.72\%$	$21.9/103.3 = 21.20\%$	$23.7/127.6 = 18.57\%$	$23.5/252.5 = 9.31\%$	$-3.7/246.8 = -1.50\%$
Hornby Group	$-3.4/37 = -9.19\%$	$1.4/38.3 = 3.66\%$	$1.5/43.1 = 3.48\%$	$-5.9/36 = -16.39\%$	$-12.1/24.3 = -49.79\%$



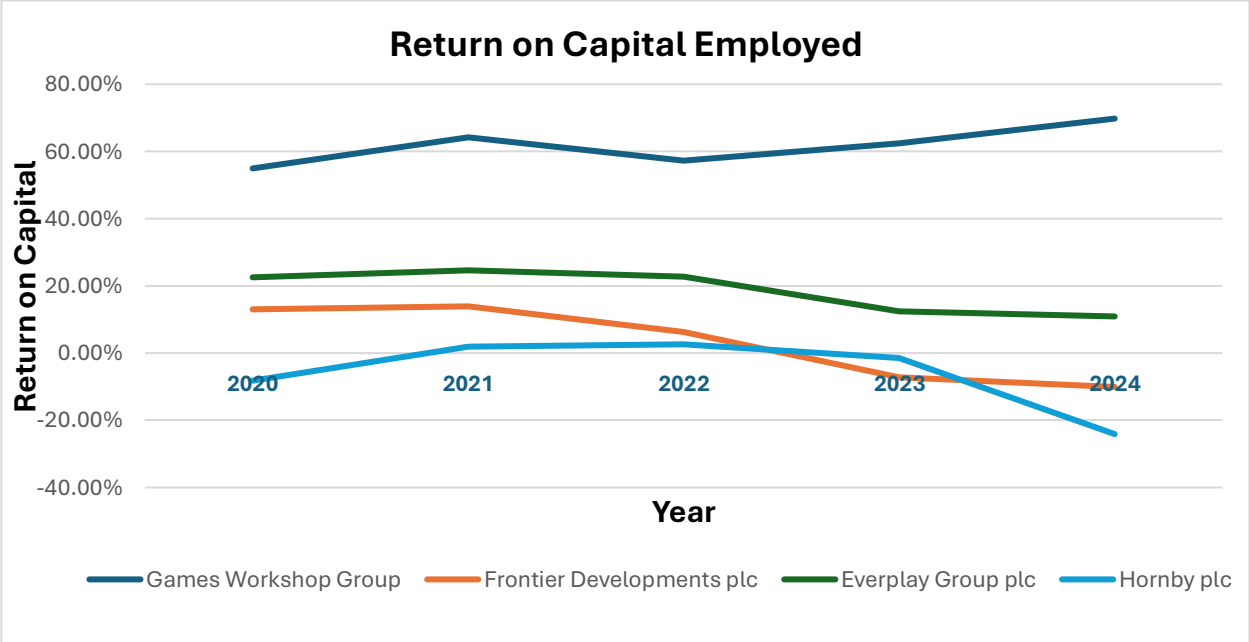
GAW's ROE is consistently and significantly higher than comparable peers for the period between 2020–2024. For example, GAW reported returns on equity of 53.33% in 2020, which increased further to 62.15% in 2021 before moderating slightly to 60.10% in 2024. In contrast, FD's performance showed a peak ROE of 19.08% in 2020 before declining sharply into negative figures, reaching -27.99% by 2024, thereby destroying shareholder value. Similarly, Everplay posted a peak ROE of 21.2% in 2021 but fell into negative territory in 2024 (-1.5%). Hornby plc consistently underperformed across the same period, with its ROE deteriorating further to -49.79% in 2024.

Return on Capital (ROC)

Return on Capital (ROC) calculated as, $((Earnings\ Before\ Intrest\ and\ Tax/Capital\ Employed)(Capital\ Employed = Total\ Assets - Total\ Current\ Liabilities))$ broadens the scope of analysis to include the return generated on all long-term capital invested in the business, including both debt and equity. It is a measure of the efficiency of capital allocation.

Table 3: Return on capital employed of GAW and its peers over the period 2020-2024

	2020	2021	2022	2023	2024
GAW	90.3 $/(207.6-43.3)$ $=54.96\%$	152.1 $/(281.7-44.7)$ $=64.18\%$	158.2 $/(321.1-44.6)$ $=57.22\%$	173.8 $/(326.8-48.2)$ $=62.38\%$	204.4 $/(351.3-58.4)$ $=69.78\%$
Frontier Dev	16.6 $/(143.9-16.4)$ $=13.02\%$	19.9 $/(161.5-18.4)$ $=13.91\%$	8.9 $/(169.6-25.7)$ $=6.18\%$	-8.5 $/(141-22.4)$ $=-7.17\%$	-10.1 $/(117.4-17.2)$ $=-10.08\%$
Everplay Group	19.1 $/(96.5-11.9)$ $=22.58\%$	26.3 $/(124.2-17.4)$ $=24.63\%$	30.4 $/(158.6-25.3)$ $=22.81\%$	$34/(327.7-54)$ $=12.42\%$	28.2 $/(297.7-39.5)$ $=10.9\%$
Hornby Group	-3.2 $/(44.7-5.3)$ $=-8.12\%$	$0.8/(48.9-8)=$ 1.96%	$1.2/(53.8-7.9)$ $=2.61\%$	-0.6 $/(54.2-15.8)$ $=-1.56\%$	-6.7 $/(55.1-27.3)$ $=-24.10\%$



Games Workshop's ROC narrative is consistent with its other profitability metrics. The company consistently generates a return on its total capital base of around 40%. This indicates that every pound of long-term capital invested in the business whether from lenders or shareholders generates a very high return.

The peers, by contrast, struggle to generate adequate returns on their capital. Frontier and Hornby have seen their ROC turn negative, meaning they are destroying capital value. Everplay's ROC has been on a downward trend, halving from its 2021 peak to just 10.9% in 2024.

Profitability Summary: Across all major profitability indicators, Games Workshop consistently outperforms its industry peers by a significant margin. This outperformance does not appear to be cyclical or driven by short-term factors; rather, it reflects a sustained pattern of superior value creation over time.

Margin Analysis:

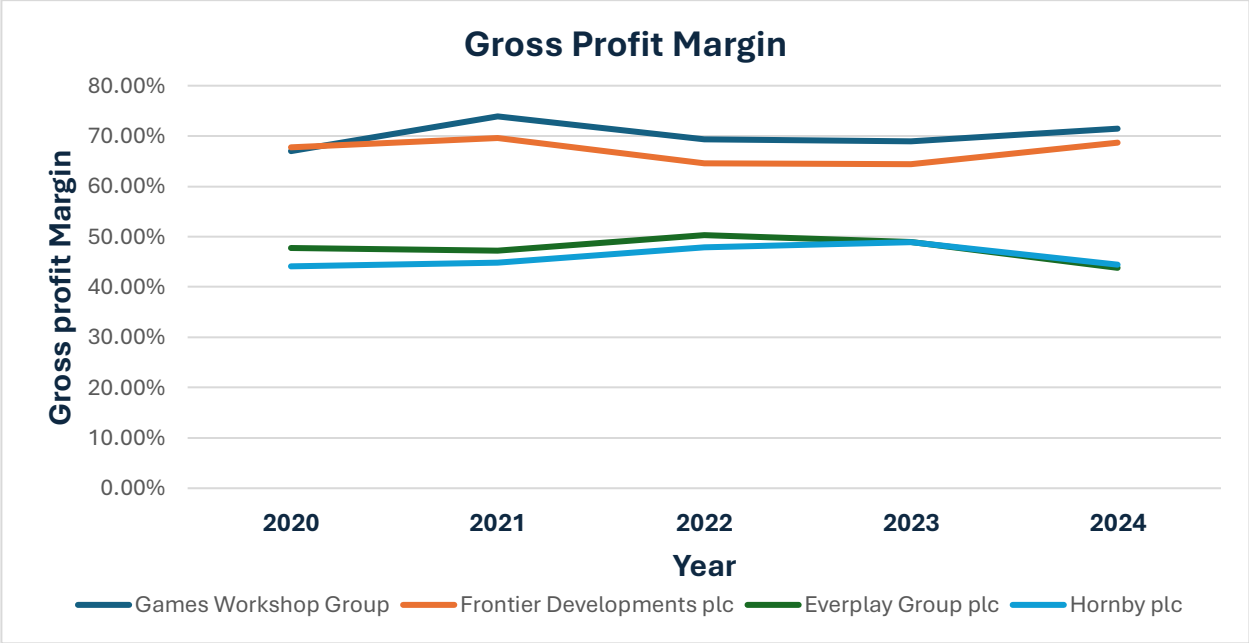
Margin analysis assesses how effectively revenue is converted into gross profit, providing a direct lens on cost control and operating efficiency. In Games Workshop's case, the profitability profile is anchored by strong and notably stable margins, which help to sustain elevated returns over time.

Gross Profit Margin

The Gross Profit Margin $\left((Revenue - COGS)/Revenue\right)$ is the measure of the basic profit margins of the companies' products and indicates the pricing power and efficiency of the company's production.

Table 4: Gross Profit Margin of GAW and Its peers for the period 2020-2024:

	2020	2021	2022	2023	2024
GAW	180.6/269.7 = 66.96%	273.2/369.5 = 73.9%	287.4/414.8 = 69.3%	324.5/470.8 = 68.9%	375.4/525.7 = 71.4%
Frontier Dev	51.6/76.1 = 67.8%	63.2/90.7 = 69.7%	73.6/114 = 64.6%	67.3/104.6 = 64.3%	61.3/89.3 = 68.7%
Everplay Group	29.5/61.8 = 47.7%	39.1/83 = 47.1%	45.5/90.5 = 50.3%	69.6/142.3 = 48.9%	69.6/159.1 = 43.8%
Hornby Group	16.7/37.8 = 44.2%	21.8/48.5 = 44.9%	25.7/53.7 = 47.9%	26.9/55.1 = 48.8%	25/56.2 = 44.5%



Games Workshop has consistently demonstrated superior profitability relative to its peers, achieving a record-high gross profit margin of 73.9% in 2021 and maintaining a level above 70% thereafter. Even within an inflationary environment, Games Workshop has preserved high margins, reporting 71.4% in 2024, which indicates effective cost management and the ability to implement price adjustments without eroding demand.

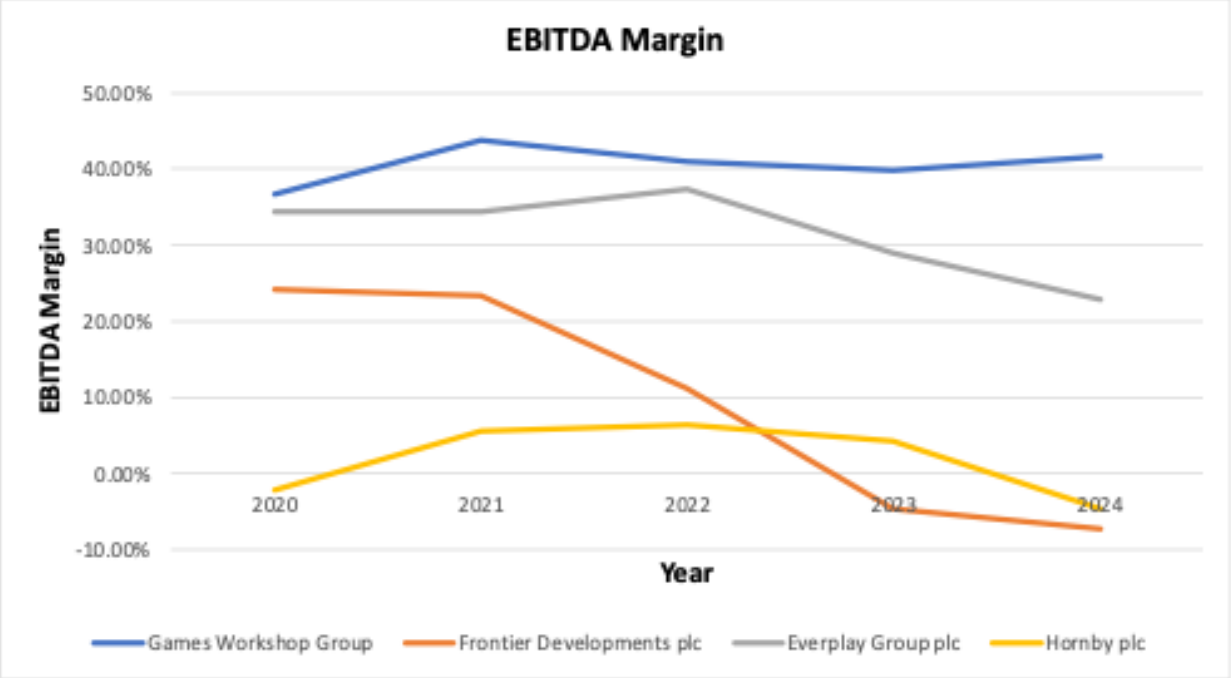
In contrast, peer companies such as Frontier Developments, Everplay, and Hornby operate with lower or more volatile gross margins. Frontier Developments, while benefitting from the relatively low marginal costs characteristic of software production, has reported margins that fluctuate and remain below those of Games Workshop in recent years. Everplay and Hornby generally report gross margins within the 40–50% range, reflecting both more competitive market conditions and weaker pricing power. In this context, Games Workshop's consistently elevated margins highlight the relative strength of its business model and its capacity to generate superior returns compared to industry counterparts.

Earnings Before Interest, Tax, Depreciation and Amortization(EBITDA) Margin

The EBITDA Margin ($EBITDA/Revenue$)gives a clearer picture of operating based profitability as the calculations exclude non-cash expenses like depreciation and amortization.

Table 5: EBITDA margin of GAW and its peers

	2020	2021	2022	2023	2024
GAW	99.1/269.7= 36.7%	161.3/369.5= 43.7%	169.9/414.8= 40.96%	187.5/470.8= 39.8%	218.8/525.7= 41.6%
Frontier Dev	18.3/76.1 = 24.05%	21.1/90.7 = 23.3%	12.6/114 = 11.05%	-4.8/104.6 = −4.6%	−6.5/89.3 = −7.3%
Everplay Group	21.1/61.8 = 34.3%	28.5/83 = 34.3%	33.8/90.5 = 37.4%	41.3/142.3 = 29.02%	36.3/159.1 = 22.8%
Hornby Group	−0.8/37.8 = −2.1%	2.7/48.5 = 5.6%	3.4/53.7 = 6.3%	2. 3/55.1 = 4.2%	−2.6/56.2 = −4.6%



With the high gross margin of Games Workshop, there is a high proportion they can convert into operating profit. Hence, Games Workshop delivered a very high EBITDA margin of 43.7% in 2021. Taken alongside operating expenses, this points to effective control of SG&A rather than gross margin expansion alone. The SG&A ratio, in the 31–

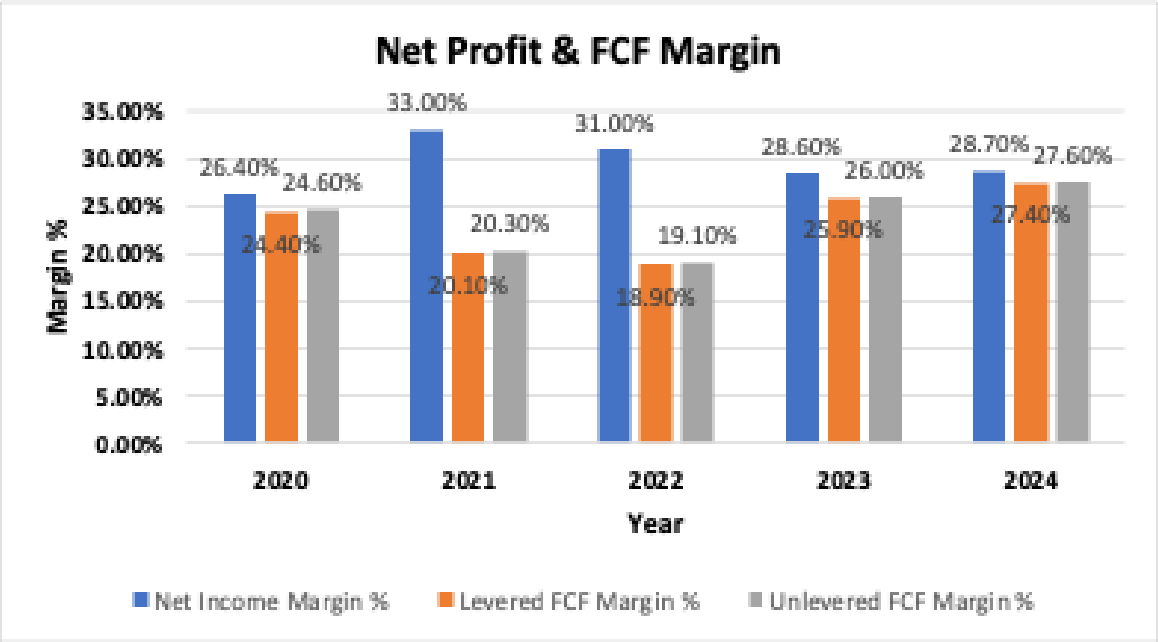
39% range, reflects ongoing investment in retail and marketing while remaining sufficiently contained to support a healthy operating margin. Frontier's EBITDA margin has gone negative, so its OpEx are overhead to gross profit. Everplay has decent margin but trending downwards. Hornby's operating profitability is minimal at best, then negative in 2024.

Net Profit Margin & Free Cash flow Margins

The Net Profit Margin ($Net\ Income\ /\ Revenue$) is the bottom line, it tells us what percentage of revenue is left over after every other expense has been accounted for. The dataset also has Levered and Unlevered Free Cash Flow margins which is equally important to judge cash generated profitability.

Table 6: Net Profit margin of GAW over the period 2020-2024

	2020	2021	2022	2023	2024
GAW	$71.3/269.7 = 26.44\%$	$122/369.5 = 33.02\%$	$128.4/414.8 = 30.95\%$	$134.7/470.8 = 28.61\%$	$151.1/525.7 = 28.74\%$



Games Workshop has a consistently strong Net Profit margin, high 20s to low 30s. The level of net profit is extremely rare and sought after. Essentially it indicates the company is a cash generating machine. More importantly, extraordinary cash flows supporting accounting profits.

The Levered Free Cash Flow (FCF) Margin shows the portion of revenues transformed into cash available to equity holders after covering all operating expenses and obligations for debts. The Unlevered FCF Margin shows the cash available to all capital holders, including debt holders and equity holders. Games Workshop runs solid FCF margins with huge margins consistently close to its profits and hence are very high quality earnings. This shows that the profits are actual cash, not profits on paper. The FCF margins for 2023, and 2024, are exceptionally strong, showing especially strong FCF margins of 27.4% in 2024 levered and 27.6% unlevered, showing superb cash generation.

Efficiency Analysis:

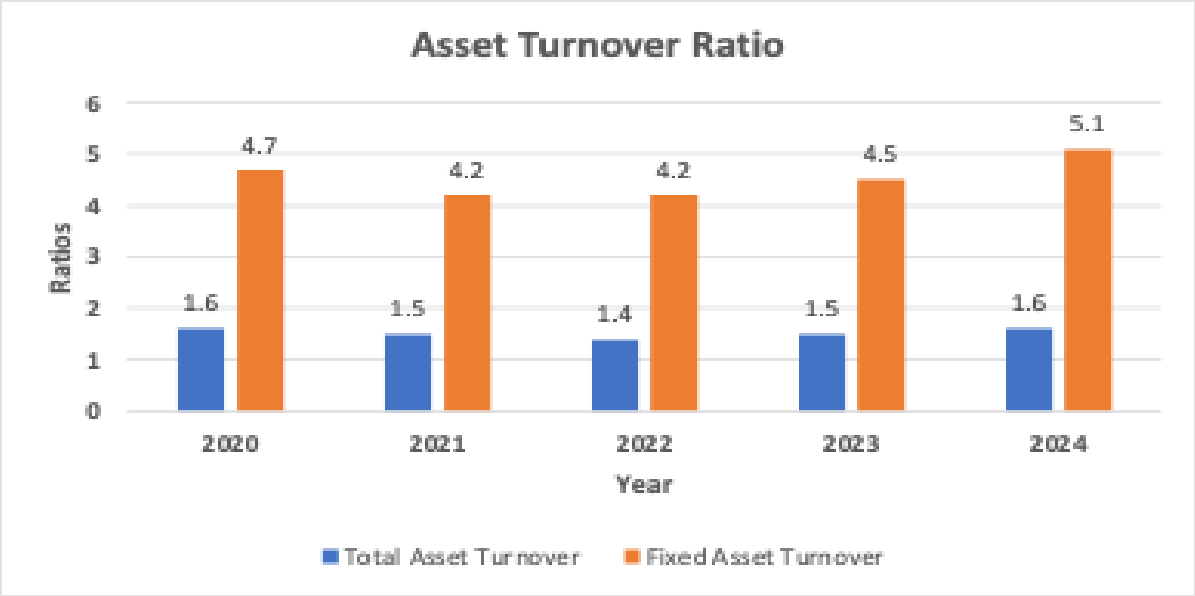
Efficiency ratios measure how effectively a company used its assets (and therefore also manages its liabilities) to generate sales and cash. Operational efficiency is an important pillar of financial performance for Games Workshop.

Total Asset Turnover = Net Sales ÷ Average Total Assets

Fixed Asset Turnover = Net Sales ÷ Average Net Fixed Assets

Table 7: Total asset turnover and fixed asset turnover ratios of GAW for the period 2020-2024.

GAW	2020	2021	2022	2023	2024
Total Asset Turnover	$269.7/172.6 = 1.6x$	$369.5/244.65 = 1.5x$	$414.8/301.4 = 1.4x$	$470.8/323.95 = 1.5xx$	$525.7/339.05 = 1.6x$
Fixed Asset Turnover	$269.7/57.05 = 4.7x$	$369.5/87.3 = 4.2x$	$414.8/99.45 =$ 4.2x	$470.8/103.85 = 4.5x$	$525.7/103.6 = 5.1x$



Games Workshop's Total Asset Turnover is quite steady and strong, coming restfully with low consistency at 1.5x for years on ending. This means for every pound of assets, revenues equals £1.50 for dollars of asset. That is a creditable level of efficiency and very strong ROA (ROA = Profit Margin Total Asset Turnover) given their solid margins.

Games Workshop's Fixed Asset Turnover is even better, where they increased from 4.2x in 2022 to 5.1x in 2024. This indicates the company was almost twice as efficient at using its property, plant and equipment as it was the preceding year to generate sales, which indicates rising efficiency, and provides a positive sign of expectation for future profitability.

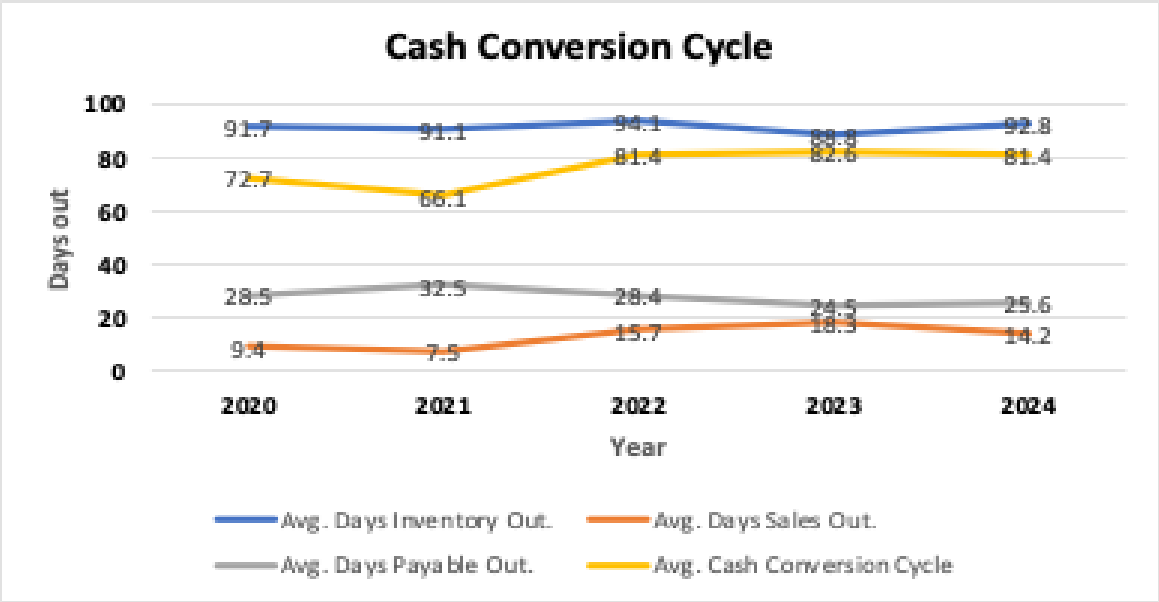
Working Capital Management:

The Cash Conversion Cycle (CCC) measures the amount of days it takes for a company to convert its investments made in inventory and other resources into cash that can be generated from sales. The shorter the cycle the better.

The CCC is determined as Days Inventory outstanding (DIO) + Days Sales Outstanding (DSO) - Days Payable Outstanding (DPO).

Table 8: Cash Conversion Cycle of Games Workshop over the period 2020-2024.

GAW	2020	2021	2022	2023	2024
Cash Conversion Cycle	$91.7 + 9.4 - 28.5 = 72.7$	$91.1 + 7.5 - 28.4 = 66.1$	$94.1 + 15.7 - 28.4 = 81.4$	$88.8 + 18.3 - 24.5 = 82.6$	$92.8 + 14.2 - 25.6 = 81.4$



Games Workshop maintains a very healthy and well-managed Cash Conversion Cycle.

- **Days Inventory Outstanding** ($DIO = \frac{Average\ Inventory}{COGS} \times 365$) is stable at around 90 days. For a manufacturing company with a vast product range, this is a reasonable and well-controlled figure. The inventory turnover of ~4.0x confirms this steady management.
- **Days Sales Outstanding** ($DSO = \frac{Average\ Accounts\ Receivable}{Credit\ Sales} \times 365$) is very low, typically under 20 days and often much lower. This is excellent, suggesting most of its sales are cash-based (likely from its own retail stores and online) or that its wholesale customers pay very promptly. This is reflected in its high Accounts Receivable Turnover (e.g., 26.2x in 2024).
- **Days Payable Outstanding** ($DPO = \frac{Average\ Accounts\ Payable}{Credit\ Sales} \times 365$) is around 25-30 days, which is a reasonable period to pay its suppliers.

The resulting CCC is consistently below 85 days, and was as low as 66 days in 2021. This indicates a high degree of operational efficiency. The company converts its working capital into cash in a relatively short period of time, which supports its liquidity and reduces the need for external financing.

Liquidity Analysis:

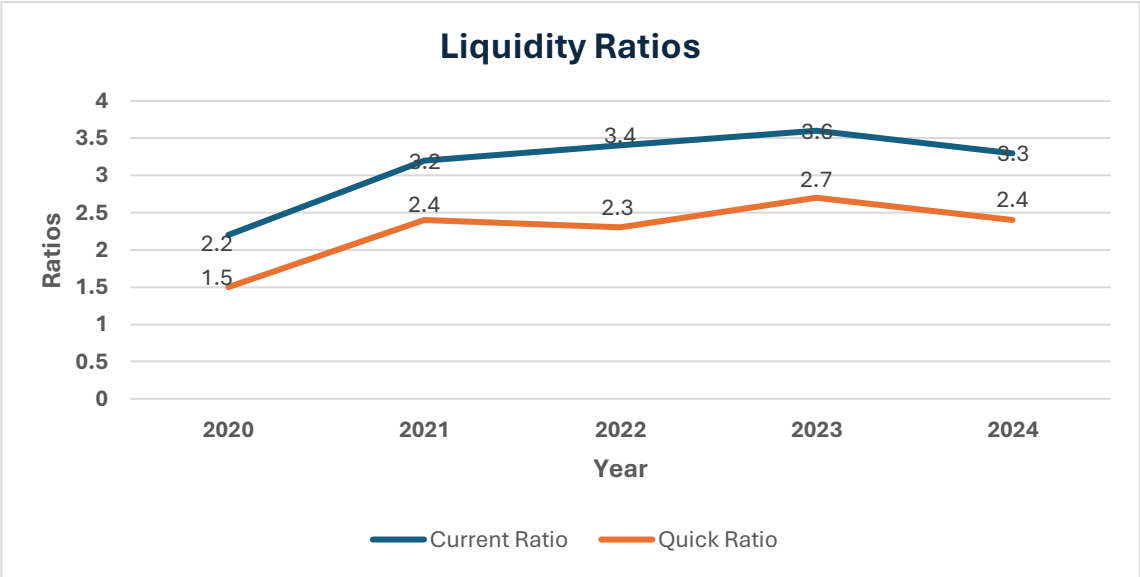
Liquidity reflects a firm's capacity to meet short-term obligations.

Current Ratio: $Current\ Assets / Current\ Liabilities$

Quick Ratio: $\boldsymbol{C}urrent\ Assets - Inventory - Prepaid\ Expenses / Current\ Liabilities$

Table 9**:** Current Ratio and Quick ratio of Games Workshop over the period 2020-2024

GAW	2020	2021	2022	2023	2024
Current Ratio	93.4/73.9 = 2.2	144.4/85.4 = 3.2	153.8/86.4 = 3.4	174/91.7 = 3.6	191.9/99.9 = 3.3
Quick Ratio	(93.4 − 20.7 − 6.7)/43.3 = 1.5	(144.4 − 27.5 − 9.9)/44.7 = 2.4	(153.8 − 38.4 − 11.7)/44.6 = 2.3	(174 − 33 − 13.2)/48.2 = 2.7	(191.9 − 42.2 − 12.1)/58.4 = 2.4



Games Workshop's position is very strong, with the current ratio consistently above 2.0x and rising to around 3.3x in 2024, implying £3.30 of current assets per £1 of current liabilities. The quick ratio is 2.4x in 2024 indicating that short-term liabilities can be covered more than twice over without relying on inventory. Together, these metrics point to high-quality liquidity and a substantial buffer against near-term cash flow risk.

Solvency Analysis:

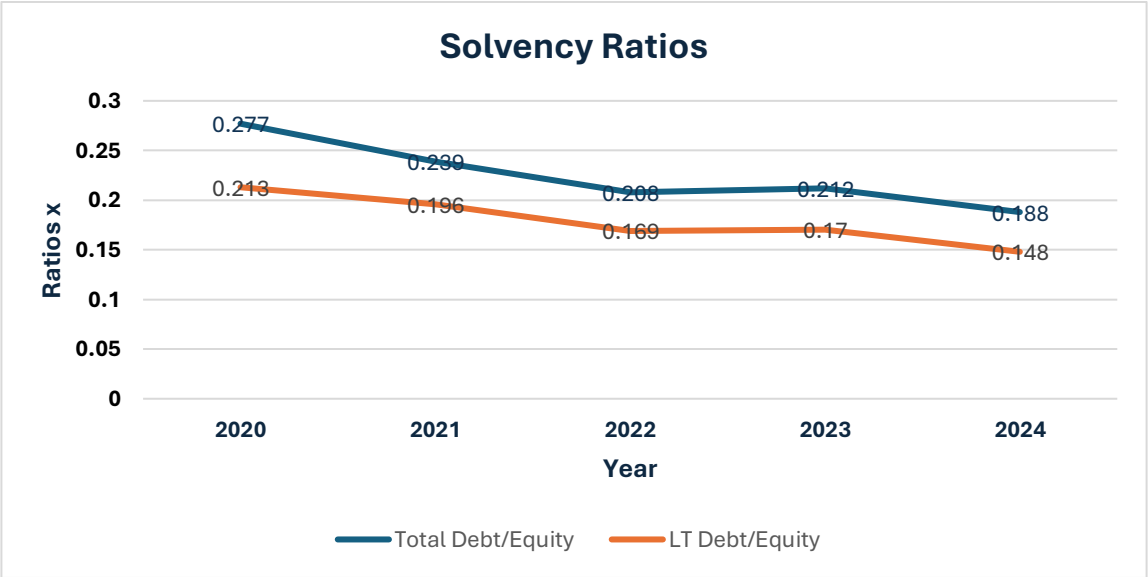
Solvency ratios assess a company's long-term financial health and its ability to manage its debt burden. Games Workshop's conservative approach to financing is a key element of its low-risk profile.

Total Debt to Equity= Total Debt/ Total Equity

LT Debt to Equity= Long Term Debt/Total Equity

Table 10: Total Debt to Equity and Long Term Debt to equity of Games Workshop over the period 2020-2024.

GAW	2020	2021	2022	2023	2024
Total Debt/Equity	37/133.7 = 0.277	47/196.3 = 0.239	48.9/234.7 = 0.208	49.9/235.1 = 0.212	47.2/251.4 = . 0.188
LT Debt/Equity	28.5/133.7 = 0.213	38.4/196.3 = 0.196	39.7/234.7 = 0.169	40/235.1 = 0.170	37.2/251.4 = 0.148



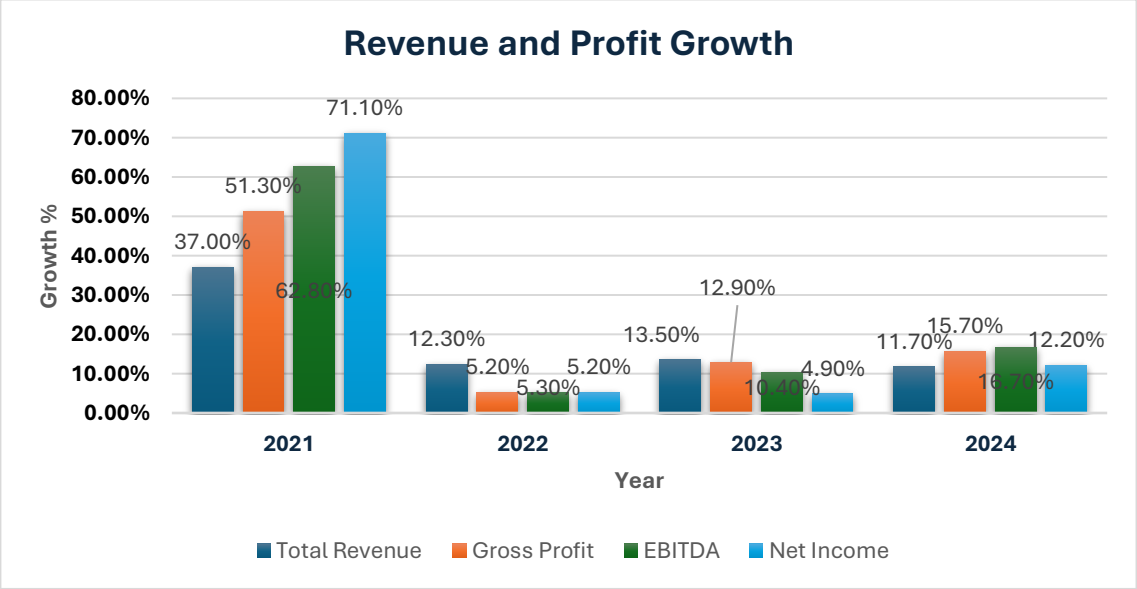
Games Workshop operates with a very low level of debt. Its Total Debt-to-Equity ratio has been on a downward trend, falling from 0.277 in 2020 to just 0.188 in 2024. This

means that its debt is less than 19% of its equity financing. This minimal reliance on debt means the company has very low financial risk. It is not beholden to lenders, and its profits are not significantly eroded by interest payments. This provides immense strategic flexibility to invest in growth, withstand economic downturns, and consistently pay dividends.

Growth Analysis:

Growth metrics provide a dynamic picture of a company's performance. Games Workshop has delivered impressive and consistent growth across its top and bottom lines.

Revenue and Profit Growth



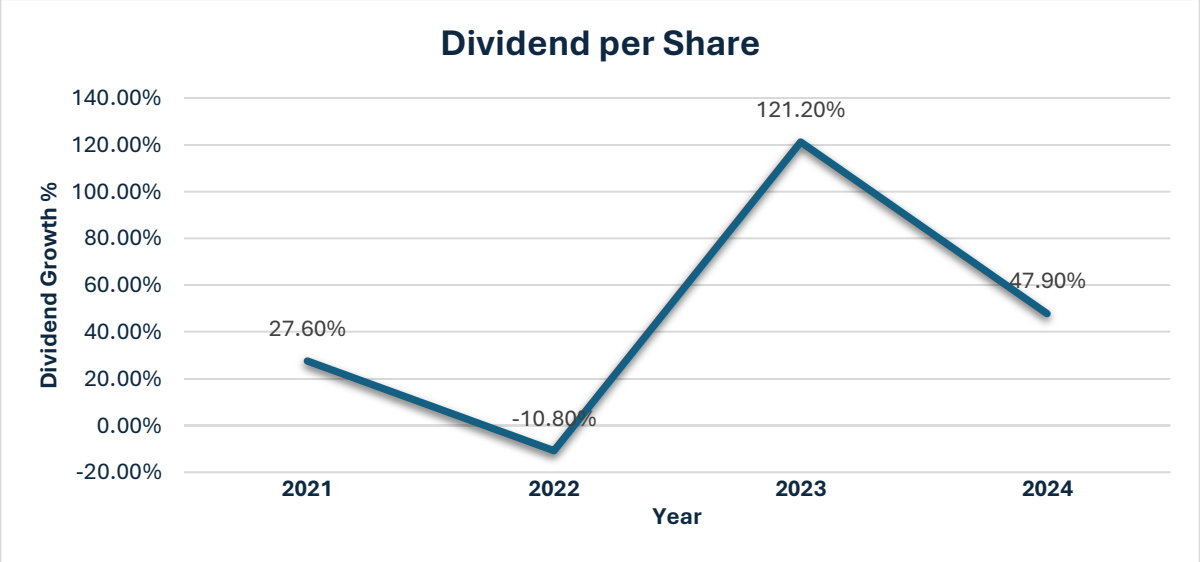
Following the lockdown-driven surge in 2021, Games Workshop has normalised to a more sustainable yet strong double-digit growth path, with revenue growth consistently above 11%. Notably, profitability has outpaced top-line expansion in 2024 (e.g., EBITDA growth of 16.7% versus revenue growth of 11.7%), indicating

margin expansion and rising operating leverage. This pattern is indicative of improving unit economics and is a positive signal for earnings durability.

Asset and Equity Growth

The data shows that Games Workshop's growth has been well-managed. The Common Equity growth (e.g., 19.6% in 2022) is strong, driven by retained earnings. The growth in Total Assets has been managed effectively, leading to the high asset turnover rates previously discussed.

Dividend Growth



Games Workshop has a policy of returning "truly surplus cash" to shareholders. This leads to fluctuating dividend growth, but the overall trend is strongly positive. The massive 121.2% increase in 2023 and 47.9% increase in 2024 are testaments to its immense cash-generating capabilities and commitment to shareholder returns. This is a direct result of the high free cash flow margins analyzed earlier. None of the peers listed show any dividend payments, further distinguishing Games Workshop as a shareholder-friendly company.

Integrated Analysis and Synthesis

In 2024, Games Workshop reported a net profit margin of 28.7%, reflecting the company's capacity to generate earnings relative to sales. Its total asset turnover was 1.6x, indicating the level of sales generated per unit of assets employed. The equity multiplier, calculated as total assets divided by total equity, was 1.40x in 2024. This was derived from liabilities representing 28.4% of total assets and an equity ratio of 71.6%, resulting in low financial leverage. Taken together, these measures demonstrate how profitability, efficiency, and leverage combine to shape the firm's overall return on equity within a mutually reinforcing structure.

Therefore, the 62.1% ROE reported in 2024 is best attributed to operational profitability and efficiency rather than balance-sheet leverage. This pattern is indicative of a high-quality and relatively low-risk business model, with returns grounded in core economics rather than financial gearing. In short, the evidence points to sustained operating excellence."

A Virtuous Cycle**:** The analysis reveals a virtuous cycle in action.

1. Strong IP and brand allow for high gross margins.
2. Efficient operations and cost control translate these into high net profit and FCF margins.
3. Strong cash flow funds investment in growth (capex growth was positive but managed), R&D, and marketing, further strengthening the brand, while also allowing for generous dividend payments.
4. This organic cash generation eliminates the need for significant debt, leading to a low-risk, fortress-like balance sheet (low solvency risk and high liquidity).
5. This financial strength and consistent growth drive an exceptional and high-quality return on equity, creating immense shareholder value.

Financial Strengths and Weaknesses:

- **Overwhelming Strengths:** The list is long: unparalleled profitability, industry-leading margins, robust cash flow, high operational efficiency, an untarnished balance sheet, and consistent, profitable growth.
- **Potential Weaknesses/Risksn(not directly evident from ratios):** The primary weakness is not financial but strategic: a high degree of dependence on its core Warhammer IP. While this IP is immensely strong, any decline in its popularity would have a significant impact. The financial ratios, however, show no signs of this risk materializing during the analysis period. The business is a financial fortress.

Industry Benchmark Analysis

This part of chapter 2 benchmarks the financial performance of Games Workshop Group (GAW) against three UK listed peers over the most recent five-year period (FY2020–FY2024). Five-year arithmetic means were calculated for each ratio for GAW and for each peer. An equal-weight peer benchmark (simple average of the three peers' five-year means) was then derived. Where the peer distribution was highly skewed, the median of the peer means is also reported. Standard deviations for GAW indicate the stability of each ratio across time.

Benchmark results

Ratio	GAW – 5-yr mean	GAW median	GAW std	Peer – 5-yr mean	Peer median	Peer std
Return on Assets (%)	**35.1%**	33.5%	2.6%	0.1%	2.5%	2.3%
Return on Equity (%)	**62.5%**	59.6%	5.9%	1.5%	0.6%	11.2%
Gross margin (%)	**70.1%**	69.3%	2.4%	53.6%	47.6%	9.6%
EBITDA margin (%)	**40.6%**	41.0%	2.3%	14.3%	9.3%	12.6%
Net income margin (%)	**29.5%**	28.7%	2.3%	**–7.1%**	–7.1%	0.0%
Current ratio (x)	**3.14x**	3.30x	0.49x	3.09x	3.10x	0.03x
Quick ratio (x)	**2.26x**	2.40x	0.40x	2.34x	2.84x	0.85x
Debt/equity (x)	**0.23x**	0.21x	0.03x	0.16x	0.22x	0.10x
Inventory turnover (x)	**4.0x**	4.0x	0.06x	41.8x	81.9x	40.1x
Accounts receivable turnover (x)	**31.2x**	26.2x	10.6x	20.0x	9.1x	15.7x
Cash conversion cycle (days)	**76.8**	81.4	6.4	114.3	221.7	107.5

Critical interpretation

Profitability. GAW exhibits markedly superior profitability across all metrics. Average return on assets (35.1%) and return on equity (62.5%) far exceed the peer benchmark (ROA near zero and ROE 1–2%), indicating that GAW generates significantly higher

returns from both assets and equity. Its gross margin of around 70.1% and EBITDA margin of 40.6% confirm the source of this outperformance—strong pricing power, an attractive product mix and efficient cost control. Net margin (29.5%) also dwarfs the negative peer average. This points to a structurally more profitable business model, supported by proprietary intellectual property and a direct, high-margin retail channel.

Liquidity. GAW maintains a current ratio of 3.1x and quick ratio of 2.3x, broadly in line with the peer benchmark. This reflects comfortable short-term liquidity and operational flexibility. Given the high cash generation noted above, these ratios also suggest an opportunity for capital reallocation (e.g. higher dividends or reinvestment) rather than an immediate need to strengthen liquidity.

Leverage. With debt-to-equity of only 0.23x, GAW operates with modest leverage, similar to its peers. High profitability is therefore achieved without significant financial risk, which strengthens the sustainability of returns on equity and offers capacity for future investment or acquisitions.

Efficiency and working capital. GAW's inventory turnover averages 4x, far below the peer mean of 42x; however, the peer benchmark here is distorted by extreme values from firms with minimal physical stock. In such cases, the peer median and qualitative differences in business models provide a more meaningful comparison. GAW's receivables turnover (31x) and cash conversion cycle (77 days) show efficient collection practices and a reasonable working-capital commitment for a product-based retailer. Nevertheless, the longer holding of inventory compared with digital or distribution-based peers can tie up capital and poses a risk in periods of demand weakness.

Volatility and comparability. GAW's ratios show low standard deviations over the five years, indicating consistent performance. In contrast, peer ratios—particularly profitability and working-capital metrics are highly dispersed. This heterogeneity limits the reliability of cross-company averages for certain metrics and underlines the importance of presenting both mean and median figures, as done here.

Synthesis

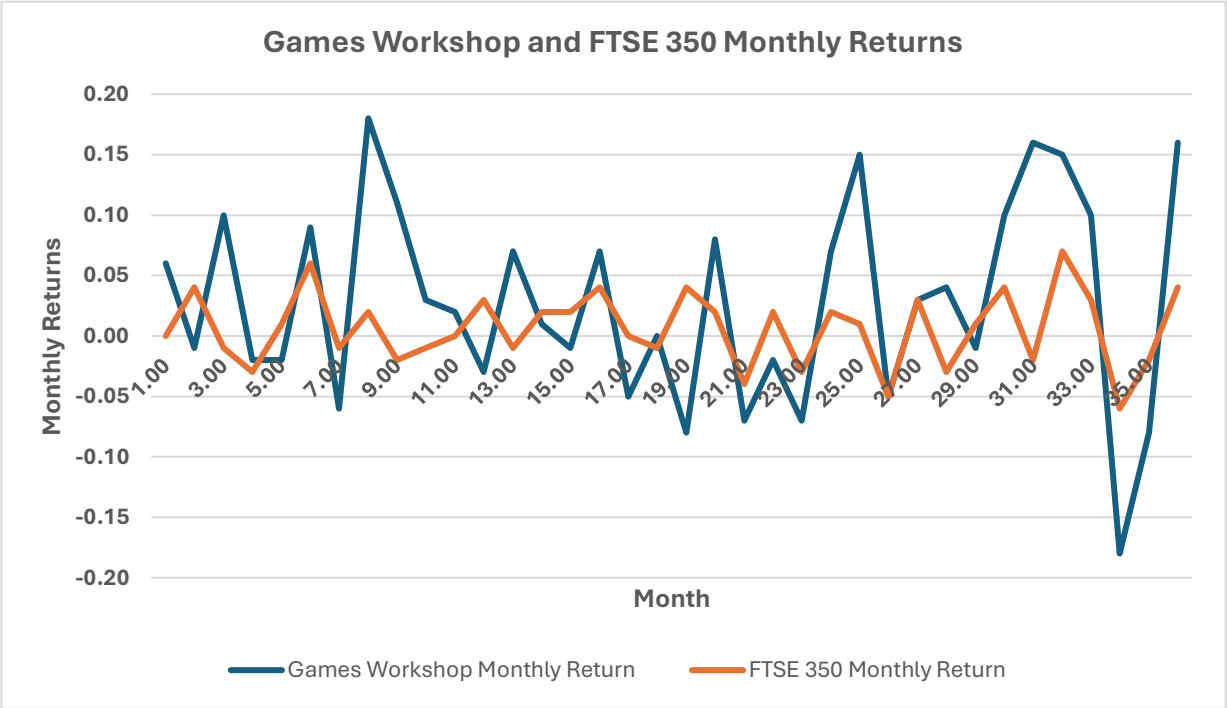
Overall, Games Workshop demonstrates a clear competitive advantage in profitability and return generation, achieved with conservative leverage and solid liquidity. These features support a premium valuation and provide headroom for continued investment. The main area of relative weakness is working-capital intensity, but direct peer comparisons are limited because of the nicheness of its industry.

Abnormal Buy and Hold Return (ABHR) Analysis

This section presents the analysis of the Abnormal Buy and Hold Return (ABHR) for Games Workshop over the observed period. The ABHR measures the performance of the stock relative to a benchmark, in this case, the FTSE 350 index, by comparing the cumulative return of the stock with that of the benchmark.

The primary benchmark is the FTSE 350 Total Return index, which aggregates the FTSE 100 and FTSE 250 constituents and therefore spans Games Workshop's transition from FTSE 250 to FTSE 100 following the December 2024 FTSE Russell quarterly review. This avoids benchmark breaks due to membership changes and provides a stable UK market baseline.

Data Overview



The dataset consists of monthly returns for Games Workshop and the FTSE 350 index over a sample period of 36 months obtained from yahoo finance. The Games Workshop monthly returns vary significantly, ranging from a minimum of **-0.18** to a maximum of **+0.18**, indicating periods of both underperformance and outperformance relative to the benchmark.

Abnormal Return Calculation

The **Abnormal Buy and Hold Return (ABHR)** over the sample period is computed by aggregating the abnormal returns over the time horizon:

$$ABHR = \prod_{t=1}^{n}(1+r)_{\text{stock}} - \prod_{t=1}^{n}(1+r)_{\text{benchmark}}$$

$(1+r)_{stock}$ Represents the cumulative, compounded return on the specific stock being analyzed (Games Workshop Group Plc).

$(1+r)_{benchmark}$ Represents the cumulative, compounded return of FTSE 350.

Example calculation for the first 4 months:

$$(1+r_{stock\ t1})(1+r_{stock\ t2})(1+r_{stock\ t3})(r_{stock\ t4}) - (1+r_{benchmark\ t1})(1+r_{benchmark\ t2})(1+r_{benchmark\ t3})(1+r_{benchmark\ t4}) =$$

$$(1\ +\ 0.16)(1\ +(-\ 0.08))(1\ +\ (-0.18)(1\ +\ 0.10) -\ (1+0.04)(1+(-0.02))(1 + (-0.06))(1+0.03)$$

$$=\ -0.024$$

Which results in a cumulative return of approximately -2.4% after the first 4 months.

Based on the full dataset, the cumulative ABHR calculated over the 36-month period is approximately **1.2154**. This figure suggests that, over the three-year period, investing in Games Workshop produced an abnormal return of 121.54% above the benchmark return.

Interpretation of Results

The positive ABHR of **1.2154** indicates that Games Workshop outperformed the FTSE 350 index over the observation period when applying a buy and hold strategy. This performance may reflect company-specific factors, such as successful product launches or favourable market conditions, which led to positive deviations from the benchmark.

Moreover, the monthly return data exhibit significant volatility, as illustrated by notable fluctuations such as **+0.18**, **-0.18**, and other large movements in the dataset. These fluctuations suggest that while the overall performance was positive, the investment was subject to substantial risk over the sample period.

The Abnormal Buy and Hold Return analysis indicates that a buy and hold strategy for Games Workshop over the analyzed period resulted in substantial outperformance relative to the FTSE 350 benchmark. However, the observed volatility highlights the importance of considering risk factors when interpreting these results.

Conclusion

This financial performance analysis indicates that Games Workshop Group plc maintains a strong financial position. Between 2020 and 2024, the company has consistently outperformed its industry peers, establishing a reference point for profitability, operational efficiency, and conservative financial management within the wider entertainment and hobby sector.

The findings show that Games Workshop exhibits superior profitability, measured by return on assets (ROA), return on equity (ROE), and return on capital (ROC), which are underpinned by sustained profit margins and effective asset utilisation. The company's margins are supported by its intellectual property portfolio and vertically integrated business model, alongside effective working capital management that facilitates rapid conversion of earnings into cash. Its financial position is further characterised by low debt and strong liquidity, resulting in low financial risk and providing scope for both long-term investment and resilience to external shocks. In addition, the firm has consistently delivered sustainable returns on equity and distributed significant cash to shareholders through dividends, indicating a record of shareholder value creation.

These results have several implications for stakeholders. For investors, they confirm the company's status as a financially resilient growth stock with durable competitive advantages. For creditors, the low leverage and strong cash flows signal limited credit risk. For management, the outcomes reflect a successful strategic model that requires continued attention to the intellectual property base on which financial performance depends. Overall, the evidence portrays Games Workshop as a highly profitable, cash-

generating, and financially stable enterprise that has established a self-reinforcing cycle of performance.

Chapter 3: Financial Management Policies of Games Workshop Group Plc

Games Workshop Group PLC has a unique market that necessitates careful financial management in order to nurture and sustain creativity while still being profitable. This additional section will add depth to our financial policy section and is an extension of the previous analysis. Financial policies in this case refer to accounting, tax management, dividend policies, cash management policies, investment policies, and risk management policies. All the financial policies discussed, are created to create a sustainable and long-term overview which wants to focus more on organic growth, compliance, and returning value to shareholders as compared to expansion through risk or leverage. This analysis is based on the company's annual reports, strategic information, financial metrics, etc for the reporting year ending June 2, 2024. Through analyzing each of these policies, we will see how Games Workshop has leveraged its position in niche market strategy and managed risks while applying fiscal prudence, to achieve ongoing high margins and resiliency in growing a successful and sustainable business.

Accounting Policy and Revenue Recognition

Games Workshop limited accounting policy is based on the International Financial Reporting Standards (IFRS) which provides an international comparative basis for the company's operations on a global scale(Games Workshop Group PLC, Annual Report 2024,p.65). This compliance enables stakeholders to measure company performance according to its design, manufacture, and retail outlets. A particular area of interest is revenue recognition, under IFRS 15 whereby a company recognizes revenue from the core product sales of miniatures, paints, rule books, etc. when control transfers to the customer, this is usually when a product is dispatched from the warehouse or when a customer collects it from the retail outlet. This approach complements the business model's vertical structure in which to run product and sales through not just the company, but all 548 stores in the world in 2024. By limiting reliance on third parties, Games Workshop lowers disputes over revenue and increases cash flow certainty.

In addition to this, the policy differentiates between trade and retail. Trade revenues are recorded upon shipment to independent retailers and retail sales are recorded when actively sold through retail stores operated by the company. More granular records like these help provide forecasts and accurate inventory position. For instance, in the twelve months ending June 2, 2024, core revenue recorded was £494.7 million up £49.3m (10.7%) compared to the last year (2023) of £445.4 million due to demand for Warhammer 40,000 and Age of Sigmar products. Conversely licensing revenue has a more simplified path: every upfront guarantees provided for IP agreements is recorded when the agreement is made because these licensing are very high margin until it comes time to hand over the IP. The licensing revenue increased to £31 million in 2024 from £25.4 million in 2023 went up due to partnerships such as the one made with Amazon Studios for Warhammer 40,000 plans for adaptations and planned video game licenses with developers such as Frontier Developments(Games Workshop Group PLC, Annual Report 2024, p.59).

Lastly, inventory valuations is also one of the most important policies, with a conservative approach that makes a provision for obsolescence consistent with sales over the prior 12 months and refresh periods. In preparing financial statements the company uses first-in, first-out (FIFO) for inventory valuation(Games Workshop Group PLC, Annual Report 2024, Notes to Financial Statements). In 2024, the inventory provision decrease reflected slower-moving items and a gross margin of 67.5%, up from 66.1% in 2023, despite raw material costs increasing. Property, plant, and equipment, such as the injection molding tools, are depreciated in a straight-line manner over their estimated useful lives between 3-10 years, enabling effective use of capital.

At year-end, intangible assets such as the IP for the Warhammer universes are amortized as efficiently as possible, reflecting the period the IP provides economic benefits, and the underlying economic value of the IP is reviewed for impairment annually. This policy decision protects the assets from overvaluation, supported by an amortization charge of £4.2m in 2024. Overall, these policies encourage conservatism and do not use some aggressiveness, such as delaying revenue recognition to show cash flow; instead, they focus on creating cash flow. They have allowed Games Workshop to protect an operating

profit of £152.0m in 2024 at an operating margin of 30.9%, which affirms the policies have been effective.

Tax Strategy and Compliance

Games Workshop's tax strategy highlights low-risk, compliant businesses with tax contributions that represent the Company's corporate values of integrity and transparency. The strategy is published annually in conformity with the UK Finance Act 2016 and covers the group and subsidiary companies. The Company's commitment to compliance covers all regions where the Company carries on business operations, including regions such as the UK, US, Europe, and Asia, which is a similar approach to the regions of operation. The strategy intentionally steers clear of artificial tax arrangements so the Company's tax planning reflects business that is a genuine commercial activity, not a long-range avoidance plan.

The Board of Directors has oversight for tax, but the day-to-day implementation of the Company's tax strategy is delegated to the Group Finance Director. The Group Finance Director is responsible as the Senior Accounting Officer. An internal tax team, and external advisors such as Deloitte, support the Group Finance Director for technical issues such as transfer pricing and cross-border transactions. The Company has a collaborative relationship with HMRC and works to create real-time discussions with HMRC to resolve queries efficiently(Games Workshop Group PLC, Tax Strategy, 2024). Discussing tax matters is especially important as North American sales were 38% of revenue in 2024, and they are subject to the US tax code as well as UK tax regulations.

In practice, the tax strategy promotes ethical tax contributions. The Company had an effective tax rate of 25.6% for 2024 on pre-tax profits of £203 million and in turn a tax charge of £51.9 million(Games Workshop Group PLC, Annual Report 2025, p.59). The effective tax rate reflects deductions for R&D expenditures on new designs for miniatures and the Company does not utilize offshore regimes to avoid taxes or longer-term approaches to reduced tax positions. For example, the Company benefits from the UK's

Patent Box regime to reduce the rate of taxation on income derived from IP by UK residents, as long as it materially contributes to the innovation process. Similarly, Games Workshop tackles tariffs and VAT (value-added tax) by varying regional pricing and using regional supply chains to deal with events such as duties arising from the departure of the UK from the EU.

Given the Company's responsible and compliant position, reputational risk is lessened. The Company publicly disclosed information about where it makes tax payments, such as by region, in its annual reports. In 2024, the Company's processes for compliance try to consider various stakeholders and stakeholder concerns; the Company earned stable after-tax profits of £155.7 million in 2024, up 14.9% from 2023, and the Company's tax policies support continued sustainable growth, sport on a good overall ethical angle to tax compliance.

Capital Structure Policy

GAW's capital structure is conservatively managed, with operations "paid for entirely from our cash flow" and an aversion to speculative financial transactions within the treasury policy. This implies limited or no structural dependence on long-term debt financing. Public filings and investor communications emphasize cash buffers and liquidity sufficiency as primary financial constraints, rather than leverage optimization. This approach is consistent with the pecking order theory (Myers & Majluf, 1984), which posits that firms prefer internal funds first, then safe debt, and only as a last resort issue equity.

- The 2017 strategic report references centralized banking relationships, board-approved treasury policies, and an explicit ban on speculative transactions. These constitute durable governance anchors for capital-structure decisions and reflect a strong preference for predictable cash management rather than exploiting leverage or arbitrage opportunities.

- More recent annual report commentary centres on sizing the liquidity buffer to three months' cash cost of operations—raised from £80 million to £85 million by May 2025—thereby setting an internal liquidity floor and de facto reducing pressure to employ debt to smooth temporary cash shortfalls. This is consistent with Myers and Majluf's (1984) view that firms with ample internal funds avoid issuing risky securities to finance normal operations.
- Third-party statistics show payouts comfortably covered by earnings over multi-year windows while revealing negligible buyback activity, suggesting management prefers cash dividends over leverage-supported repurchases. This stands in contrast to firms following a more traditional **trade-off theory** (Kraus & Litzenberger, 1973), which balances tax shields against expected distress costs to reach an optimal debt ratio. GAW's absence of an articulated leverage "target" and its lack of recurring buybacks signal that minimizing financial risk and preserving flexibility outweigh any potential tax benefits of debt.

In this sense, GAW's policy partly echoes the first proposition of Modigliani and Miller (1958), which asserts that under perfect capital markets a firm's value is independent of its capital structure. However, once real-world frictions such as bankruptcy costs, asymmetric information, and managerial incentives are introduced, MM's irrelevance breaks down. GAW's practice of holding significant liquidity buffers and avoiding debt issuance can be interpreted as a response to these frictions—especially asymmetric information and potential distress costs—precisely the issues highlighted in the pecking-order model.

A corollary of this stance is that shareholder returns are provided primarily via dividends rather than capital-structure engineering. Total shareholder yield metrics for the stock show a near-zero buyback yield and a total yield dominated by cash dividends, indicating that the company uses retained earnings to fund operations and distributions rather than debt. Overall, GAW's capital structure policy can be characterized as low-leverage, liquidity-first, and designed to support resilient operations rather than to optimize cost of

capital through leverage cycles, aligning closely with the pecking order theory and differing from trade-off theory implementations.

Dividend Policy and Shareholder Returns

Games Workshop's (GAW's) dividend policy is explicitly focused on providing cash distributions to shareholders while maintaining a safety net to run the business. Surplus cash is defined as the fund and reserves available in excess of the cash reserve for three months' worth of working capital, tax liabilities, and capital or profit-share obligations of more than £1 million. The cash buffer was established at £50 million in 2024 and will vary depending on cash generation expectations, reflecting the company's preference for liquidity during uncertain periods.

In 2024, total dividends paid amounted to 420 pence per share, an increase from 415 pence in 2023. (Games Workshop Group PLC, Annual Report 2024, p. 20). Over the previous five years, total dividends have grown at a compound annual growth rate of 18.5% as cash is generated from high-margin sales. This pattern is consistent with Lintner's (1956) model of dividend policy, which argues that firms smooth dividends relative to earnings and maintain a target payout ratio once profitability stabilises.

GAW's approach also aligns with the residual dividend theory (Miller and Modigliani, 1961), which suggests that firms should distribute only the cash left over after funding all positive-NPV investments. By holding a defined liquidity reserve before paying dividends, GAW effectively applies a residual dividend logic. CEO Kevin Rountree's stated avoidance of value-destroying acquisitions also reflects the agency theory perspective (Jensen, 1986): by returning excess cash to shareholders, management reduces the risk of wasteful spending or empire building.

For example, the company's operating cash flow of £196.2 million in 2024 (Games Workshop Group PLC, Annual Report 2025, p.71) permitted dividend payments without depleting reserves, even while capital expenditure reached £32.6 million. Total

shareholder return, inclusive of dividends, was 28.4% in 2024, showing that shareholders have been rewarded through both cash distributions and share price appreciation. This track record has attracted loyal long-term investors, with a market capitalisation of approximately £3.1 billion by mid-2025.

Overall, these policies are designed to reward loyalty and signal confidence in future earnings, echoing the signalling theory of dividends (Bhattacharya, 1979), which posits that sustained or increasing payouts convey private information about a firm's future prospects to the market.

Cash Management and Liquidity

The effective management of cash is of utmost importance in the context of Games Workshop's stability. The company has established policies in the area of cash management that value a debt-free balance sheet and have a high degree of liquidity. As at June 2, 2024, cash was £107.6 million and had grown 132.2% from £90.2 million in 2023, driven by strong operational inflows(Games Workshop Group PLC, Annual Report 2024, p. 60).

The company operates a rolling cash forecast in their cash management that allows it to keep track of cash position and liquidity to cover any unexpected occurrences in the normal operations of the business, such as a supply chain problem, economic pull-back, etc. The operating cash flow conversion ratio was 140% of operating profit in 2024, which illustrates the cash cycle management was accordingly well-placed.

The liquidity ratios were all supportive: the company's current ratio was 3.6 and its quick ratio was 2.8, (Games Workshop Group PLC, Annual Report 2024, Financial Statements) suggesting that it has sufficient resources to meet its short-term liabilities and that there is limited risk for liquidating its debt level, if it had any.

The liquidity policies include investing any excess cash into low-risk financial instruments, such as short-term deposits. The low-risk means significantly lower yields for only modest returns while keeping the principal intact.

Investment Policy and Capital Expenditure Policy

Investment policies prioritize internal improvements over alternative or external growth opportunities. Total capital expenditures were £32.6 million in 2024, as focused on factory expansion and necessary IT improvements which ultimately resulted in efficient production gains. The company steers away from mergers and acquisitions and reinvests in its core operating assets located in the UK for the company's headquarters.

Risk Management

The company pursues risk policies with support of commodity hedging for flavour and fairly wide foreign exchange hedging, from either a forward purchase contract or as an overarching concept. In 2024, foreign exchange forward purchases were used to mitigate any volatility from future business transactions in foreign currencies where 75% of its sales were in foreign currencies.

Integration with comprehensive Financial Results

Ultimately, all of the financial policies discussed herein have enabled overall revenue growth to the tune of £525 million in 2024, culminating in earnings per share of 473.3 pence (Games Workshop Group PLC, Annual Report 2024, p. 59). The company is still exposed to challenges like inflation, for which it has clear strategic maneuvers or implications.

To summarize, the Financial Strategies employed by the company are suitable for greater longevity, for the company as it continues to build a successful liquidity position.

Chapter 4: Valuation of Games Workshop Group Plc

DCF Valuation Models

1. Free Cash Flow to the Firm (FCFF) Valuation

Definition & Relevance:

FCFF represents the cash flow generated by a company that is available to all investors, both debt and equity holders. It is calculated after accounting for operating expenses and reinvestment in the business (capital expenditures and changes in working capital), but before interest payments. This model is particularly valuable when assessing the overall performance and valuation of the enterprise, regardless of capital structure.

Methodology:

The FCFF calculations are based on detailed forecasts from 2025E to 2029E, derived from adjusted EBIT (Earnings Before Interest and Taxes), with non-cash charges like Depreciation & Amortisation added back and capital expenditures and working capital adjustments deducted.

Calculation Method:

$$FCFF = EBIT \times (1 - \text{Tax Rate}) + \text{Depreciation and Amortization} - \text{Capital Expenditures} - \Delta \text{Working Capital}$$

Where:

- **EBIT** = Earnings Before Interest and Taxes
- **Tax Rate** = Corporate tax rate (expressed as a decimal, e.g., 25% → 0.25)
- **Depreciation & Amortization** = Non-cash charges added back
- **Capital Expenditures (CapEx)** = Investments in property, plant, equipment, etc.

- **Δ Working Capital** = Change in Net Working Capital = (Current Assets - Current Liabilities) for the current period minus that of the previous period

Unlevered Free Cash Flow (2025–2029):

- 2025E: £133.9 million
- 2026E: £146.7 million
- 2027E: £156.9 million
- 2028E: £170.4 million
- 2029E: £182.7 million

$$\text{Terminal Value (TV)} = \frac{\text{FCF}_{2029} \times (1+g)}{WACC - g}$$

$$\text{TV} = \frac{182.7 \times (1+0.025)}{0.055 - 0.025} = \frac{187.22}{0.03} \approx 6{,}240.67 \text{ (in £ millions)}$$

Revenue growth rate assumptions for the FCFF model were:

2025E: 7%

2026E: 6.45%

2027E: 7.07%

2028E: 6.55%

2029E: 6.50%

The revenue growth rate assumptions applied in the DCF model (ranging from 6% to 7% annually from 2025 to 2029) are based on historical CAGR figures and alignment with

the UK's projected GDP growth rates. While this approach aims to anchor projections in macroeconomic reality, it carries several important caveats.

The tabletop gaming market in which Games Workshop operates is relatively mature and highly niche. Historical periods of strong growth were significantly influenced by temporary factors such as increased consumer hobby engagement during the COVID-19 pandemic, which may not be sustainable over the long term. Industry reports suggest that after the pandemic-related surge, market expansion is expected to stabilize rather than continue at a double-digit rate.

The forecast assumes that the core Warhammer IP will continue to drive revenue growth at a consistent pace

The tax rate is assumed to remain constant at 25% throughout the forecast period. This is a reasonable simplifying assumption but ignores the potential for regulatory changes (e.g., changes to the UK Patent Box regime, alterations to corporate tax policy, or additional tariffs arising from Brexit complexities) that could materially affect future cash flows.

Key Components and Forecasted Inputs (in £m):

Year	Revenue	EBIT	D&A	Capex	Change in WC	Tax	FCFF
2025E	562.5	183.5	21.6	15.6	56.4	-25%	133.9
2026E	598.8	198.2	20.2	14.5	64.1	-25%	146.7
2027E	641.1	215.0	18.9	13.6	73.7	-25%	156.9
2028E	683.1	231.5	17.6	12.7	81.9	-25%	170.4
2029E	727.5	248.9	16.5	11.9	90.5	-25%	182.7

- **Revenue Growth**: Rises steadily from £562.5m to £727.5m over the forecast period.
- **Operating Efficiency**: Margins remain strong, and EBIT as a percentage of revenue is consistently above 30%.
- **Capital Expenditure**: Capital expenditures increase moderately in line with business expansion.
- **Working Capital**: Gradual release of working capital enhances FCFF.
- **Tax Rate**: Tax rate is assumed to be 25%(Corporate income tax)for the entire forecast period.

Games Workshop displays strong free cash flow generation, bolstered by a high operating margin, stable reinvestment policy, and improving working capital efficiency. FCFF increases consistently, reflecting both operational robustness and financial discipline.

2. **Free Cash Flow to Equity (FCFE) Valuation**

Definition & Use:
FCFE isolates the cash flow available to equity shareholders after accounting for capital expenditures, debt repayments, interest, and changes in working capital. This model is ideal for evaluating the company's ability to distribute excess funds to its equity holders.

Calculation Method: $FCFE = Net\ Income + Depreciation\ \&\ Amortisation - CapEx - \Delta Working\ Capital + Net\ Borrowing$

- **Net Income** = Profit after tax attributable to shareholders
- **Depreciation & Amortization** = Non-cash charges added back
- **Capital Expenditures (CapEx)** = Investments in property, plant, equipment, etc.
- **Δ Working Capital** = Change in Net Working Capital = (Current Assets - Current Liabilities) for the current period minus that of the previous period
- **Net Borrowing** = New Debt Issued minus Debt Repayments during the period

FCFE Breakdown (in £m):

Year	Net Income	D&A	CapEx	ΔWC	Net Borrowing	FCFE
2025E	138.8	22.7	(39.6)	(11.3)	3.7	114.3
2026E	146.7	23.8	(41.9)	(10.4)	2.3	120.5
2027E	154.9	25.0	(44.4)	(9.5)	0.8	126.8
2028E	163.4	26.2	(47.0)	(8.6)	(0.6)	133.4
2029E	172.3	27.5	(49.8)	(7.8)	(2.1)	140.1

- **Net Income Growth**: Increases from £138.8m to £172.3m.
- **Capital Expenditures**: Similar to FCFF, increases with expansion.
- **Working Capital**: Negative change each year (i.e., cash inflow from operations).
- **Net Borrowing**: Positive until 2026, then turns negative (deleveraging).

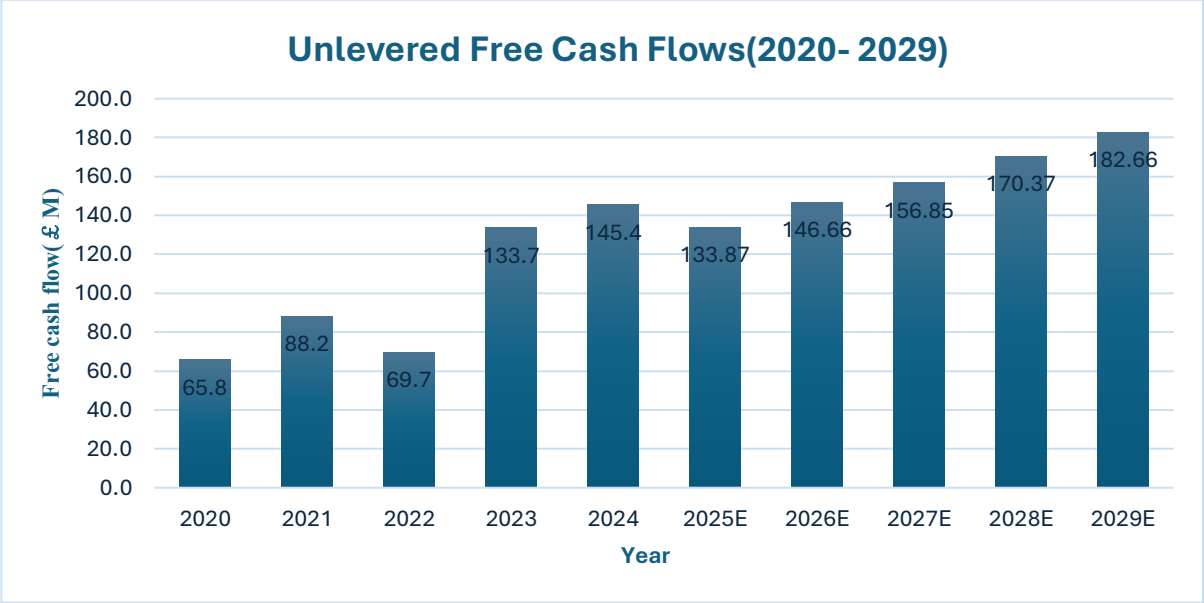
Interpretation: The FCFE approach highlights strong distributable cash flow, increasing from £114.3m to £140.1m over the forecast period. This aligns well with the company's historical dividend policy and prudent capital structure. However, the model's reliance on stable net borrowing and working capital patterns introduces risk:

- Assumes no material shifts in debt policy, capex cycles, or dividend payouts strategy, which may not hold true if GAW pursues large-scale strategic investments or faces unexpected regulatory costs.
- By focusing solely on cash available to equity, the model somewhat ignores the value created for debt holders and may understate enterprise-level risk if debt structure were to change unexpectedly.

3. Discounted Cash Flow (DCF) Valuation Based on FCFF

Overview:

The DCF method involves discounting the projected FCFF to present value using the Weighted Average Cost of Capital (WACC), which reflects the risk-adjusted required return by all investors.



Key Assumptions:

- **Risk Free Rate:** 4.63 UK 10 Year Treasury Yield was taken as the risk free rate(July 14, 2025)
- **WACC:**

  WACC = (E/V) × Re + (D/V) × Rd × (1 − Tc)

  E = market value of equity = **251.4**

  D = market value of debt = **47.2**

  V = E + D = 251.4+ 47.2 = **298.6**

R_e = cost of equity = Rf + β × (Rm − Rf) = 4.63+ 0.90 × (6.3%- 4.63%) = **6.1%**

R_d= pre-tax cost of debt = **2.75%**

t_c= corporate tax rate = **25%(UK Corporate Tax Rate)**

WACC= (251.4/298.6) × 6.1% + (47.2/298.6) × 2.75 × (1- 25%) = **5.49%**

- **Terminal Value Growth Rate(2.5%)**: Assumed based on the annual United Kingdom GDP growth rate.
- **Projection Period:** 2025E to 2029E.

DCF Components and Valuation Summary:

- **Present Value of FCFF (2025-2029):** Shown in the above chart
- **Terminal Value:** £6,261.71 million
- **Present Value of Terminal Value:** £4,793.30 million
- **Enterprise Value (EV):** £5,463.02 million
- **Add: Cash:** £107.6 million
- **Less: Debt:** £47.2 million
- **Equity Value:** £5,523.42 million
- **Shares Outstanding:** 33 million
- **Implied Share Price** $= \frac{\text{Equity Value}}{\text{Shares Outstanding}} = \frac{5{,}523.42}{33} = 167.38$ GBP
- Upside (%) $= \frac{167.38-156.00}{156.00} \times 100 = 7.3\%$

Sensitivity Analysis: To evaluate the robustness of the DCF model and understand the impact of key assumptions, a sensitivity analysis was performed. The two variables tested were:

Sensitivity Table						
		Growth Rate				
	167.38	2.00%	2.25%	2.50%	2.75%	3.00%
WACC	3.50%	340.26	404.58	501.05	661.84	983.41
	4.00%	255.06	288.85	333.92	397.00	491.63
	5.49%	145.96	155.84	167.38	181.02	197.39
	6.00%	127.30	134.61	142.96	152.60	163.84
	6.50%	113.12	118.75	125.07	132.24	140.44

The results showed that:

- A 0.5% increase in WACC reduces the enterprise value by approximately 12-15%
- A 0.25% increase in terminal growth rate raises valuation by 8-10%

This highlights the importance of accurately estimating both discount rate and growth expectations. Nonetheless, even under conservative scenarios, the valuation remained above the prevailing market price, reinforcing the undervaluation thesis.

Interpretation:

The DCF model yields an implied share price of £167.38, considerably higher than the current market price, strongly suggesting undervaluation. This is primarily driven by strong forecasted free cash flows, conservative capital expenditure, and low cost of capital (WACC of 5.49%). However, this valuation is highly sensitive to key assumptions:

- The sensitivity analysis shows valuation swings of up to 15% from minor changes in WACC or terminal growth assumptions, underlining the inherent uncertainty and the need for caution when interpreting DCF results as definitive.

Moreover, the DCF implicitly assumes that the current operational efficiency and pricing power (driven by IP strength) will sustain indefinitely, which may be overly optimistic given potential disruptive industry trends, competitor innovations, or regulatory changes.

4. Strengths and Limitations of the Models

Strengths:

- Cash flow models (FCFF/FCFE) consider full business operations
- DCF provides a dynamic, multi-factor valuation tool
- FCFE gives insights into actual distributable earnings
- Triangulation enhances reliability

Limitations:

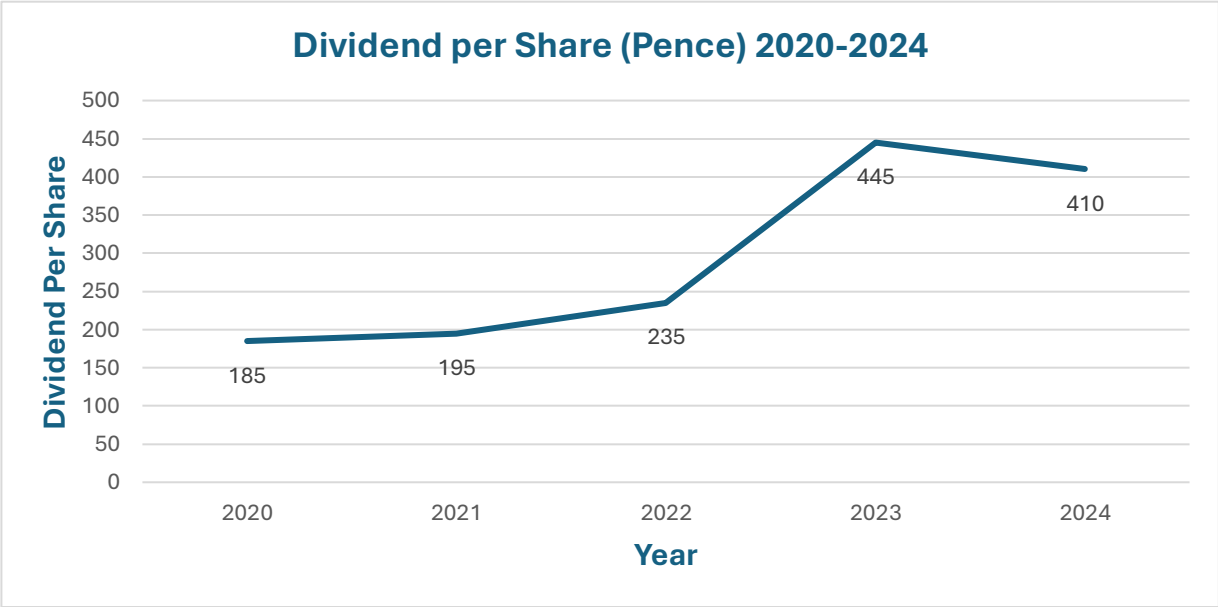
- Models are sensitive to assumptions (WACC, growth rate)
- Terminal Value often contributes over 60% to final valuation

Combining the insights from FCFF, FCFE, and DCF models, we conclude that Games Workshop is intrinsically undervalued. The company's robust free cash flow generation, conservative debt usage, and consistent operational performance support a higher fair value than reflected in current market pricing.

This reinforces the investment thesis that Games Workshop presents a compelling long-term value opportunity, especially when viewed through the lens of fundamental, cash-flow-based valuation methodologies.

Dividend Discount Model (DDM)

To estimate the intrinsic value of Games Workshop's equity, a Dividend Discount Model (DDM) was employed, which is particularly suitable for firms with a strong track record of dividend payments. Games Workshop has consistently distributed dividends, making



it a strong candidate for this valuation approach.

Methodology

The DDM assumes that the value of a stock is the present value of all future expected dividends. The model applied here uses the Gordon Growth Model (a variation of DDM), which assumes a constant dividend growth rate after a certain forecast period.

$$P_0 = \frac{D_1}{r - g}$$

Where:

- P_0 is the intrinsic value of the stock
- D_1 is the expected dividend in the next year

- r is the discount rate (cost of equity)
- g is the long-term dividend growth rate

Calculation of Intrinsic Value(2029E):

$$P_0 = \frac{487 * (1 + 0.035)}{0.0613 - 0.035} = \frac{487 * (1 + 0.035)}{0.0263} \approx 19{,}141\ pence$$

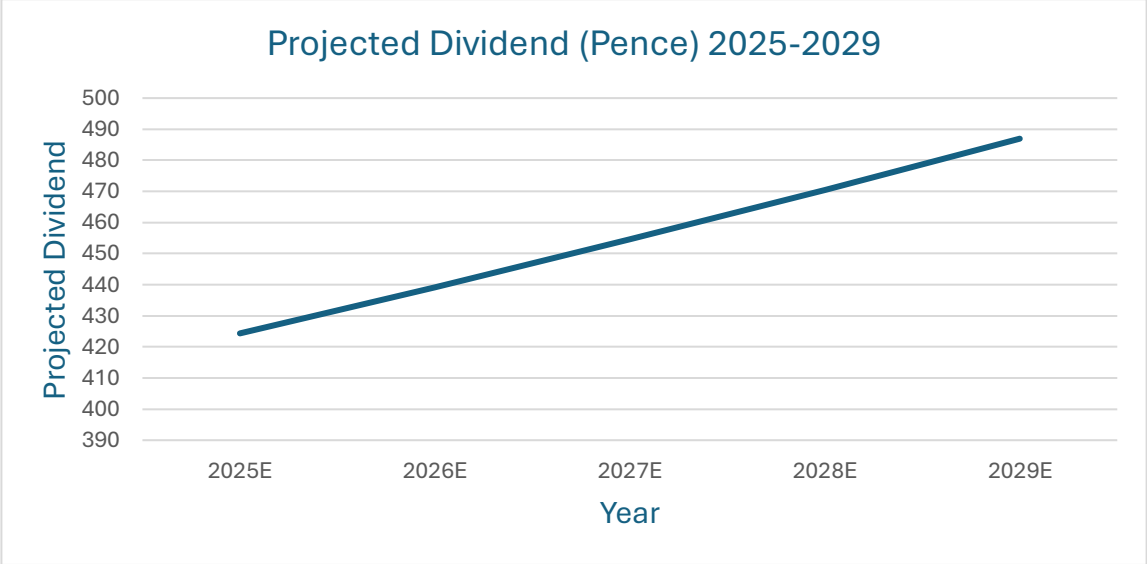
Assumptions and Inputs

- **Expected Growth Rate (g):** 3.50%. This growth rate was assumed after looking into the historical dividend growth rate of Games Workshop Group Plc and the industry average (tabletop gaming and hobby industry) dividend growth rate.
- **Discount Rate (r):** 6.13% (Cost of Equity. Calculated in the DCF model section)
- **Calculated Intrinsic Value:** 19,141 pence
- **Current Market Price (14/07/2025):** 15,600 pence
- **Implied Upside:** 22.70%

These inputs yield an intrinsic value of 19,141 pence, suggesting the stock is currently undervalued by approximately 22.7% at its current market pric

Dividend Forecast

Dividends from 2025 to 2029 were projected, incorporating actuals for past years and applying a consistent 3.0% growth rate in forward-looking estimates. The present value of forecasted dividends (2025–2029) was computed at 19141 Pence, with subsequent years contributing to the terminal value.



This strong dividend performance particularly the sharp increase in 2023 underlines Games Workshop's profitability and shareholder focus.

Sensitivity Analysis

A sensitivity analysis was conducted to observe the effect of changes in growth rate and discount rate on the estimated intrinsic value.

Sensitivity Table						
		Discount Rate				
	19,141	5.13%	5.63%	6.13%	6.63%	7.13%
Growth Rate	2.50%	18,058	15,176	13,088	11,504	10,263
	3.00%	22,952	18,593	15,626	13,475	11,845
	3.50%	30,863	23,628	19,141	16,087	13,873
	4.00%	45,788	31,769	24,322	19,703	16,559
	4.50%	84,349	47,125	32,696	25,032	20,278

From the table, it is evident that the intrinsic value is highly sensitive to both the discount rate and the assumed growth rate. At a baseline scenario of 3.5% growth and 6.13% discount rate, the model yields a fair value of 19,141 pence per share.

The Dividend Discount Model suggests that Games Workshop is undervalued based on its historical and projected dividend profile. Despite some recent volatility in dividend payouts, the company's strong brand, operational performance, and consistent shareholder returns justify the use of DDM. The sensitivity analysis further reinforces the robustness of the valuation under various economic scenarios. This model supports a bullish investment case for Games Workshop, subject to the sustainability of its dividend policy and long-term growth prospects.

This underscores the need for caution when relying on DDM outputs, especially in firms with uncertain future payout patterns or volatile earnings.

Strengths and Limitations of the Model

Strengths:

- Simple and intuitive for dividend-paying firms.
- Focuses directly on shareholder returns.
- Suitable for mature, low-growth companies with predictable payout policies.

Limitations:

- Ignores capital gains and non-dividend cash flows.
- Assumes perpetual growth beyond the forecast period, which may not reflect business cycles or competitive threats.
- Less applicable in scenarios of irregular or no dividends.

The DDM offers a structured and conservative valuation approach for Games Workshop, anchored in actual dividend behaviour and a defensible cost of equity. The model estimates an intrinsic share price significantly above the prevailing market price, suggesting a potentially attractive investment opportunity. However, this conclusion is contingent on the firm's ability to maintain dividend growth and its stable risk profile over the long term.

Present Value of Growth Opportunities(PVGO) Model

To complement the cash flow-based valuation methods (FCFF, FCFE, and DCF), we also analysed the Present Value of Growth Opportunities (PVGO) to assess the portion of the company's share price that is attributable to future growth prospects. PVGO is particularly useful in understanding how much value investors place on a company's potential to grow its earnings beyond its current capabilities.

$$PVGO = P_0 - \frac{E_1}{r}$$

PVGO is the Present Value of Growth Opportunities.

- $P_0(156)$ is the current market stock price.
- $E_1(4.84$ is the expected earnings per share (EPS) for the next period.
- $r(6.13\%)$ is the cost of equity, which is the rate of return required by equity shareholders.

$$PVGO = 156 - \frac{4.84}{0.0613} = 77.05$$

Variable	Value
Expected Growth Rate (CAGR of 3 Years)	5.49%
Expected Earnings Per Share(£)	4.84
Current Share Price(£)	156
Cost of equity	6.13%
No Growth Value Per Share(£)	78.95
PVGO	**£ 77.05**

The growth rate for the PVGO model was assumed by taking into account the CAGR of the past 3 years of the target company as well as factoring in the GDP growth rate of the UK.

Intrinsic Value Without Growth

This component estimates what Games Workshop shares would be worth today if the company generated its current earnings indefinitely, with no further growth:

This £78.95 per share represents the base intrinsic value of the company assuming a no-growth scenario. It sets a foundation on which investors would layer their expectations for future value creation.

This result indicates that approximately 49.4% of Games Workshop's share price is driven by the market's expectations of future growth opportunities — a strong endorsement of confidence in the firm's strategic prospects and ability to grow earnings..

Strategic Interpretation of PVGO for Games Workshop

The magnitude of PVGO in this context reflects:

- Strong investor confidence in Games Workshop's strategic vision and its ability to monetize existing and future intellectual property.
- The potential for expanding global presence in hobby and licensing markets.
- Continued customer loyalty and engagement with Games Workshop's community-led product ecosystems.
- IP licensing and digital platform growth as margin-enhancing opportunities.

This model demonstrates that the bulk of the company's valuation is not tied to its current earnings, but rather to its perceived ability to grow those earnings substantially in the future.

Valuation Implication

The PVGO model strengthens and complements conclusions drawn from FCFF, FCFE, and DCF models. While those models quantify value based on projected cash flows and operations, PVGO captures investor psychology and forward-looking expectations.

It confirms that Games Workshop is not merely a profitable company today, but one that markets expect to continue scaling, innovating, and expanding. This supports the thesis that Games Workshop merits a premium valuation, not only due to its current operational success but more importantly due to its future strategic growth potential.

Sensitivity Analysis – Present Value of Growth Opportunities (PVGO)

To further evaluate Games Workshop's equity value and the premium investors are willing to pay for future growth, a sensitivity analysis was conducted on the Present Value of Growth Opportunities (PVGO).

Using a base EPS of £4.84 and a cost of equity of 6.13%, the no-growth value per share was calculated at £78.95, and the market price of £156 implied a PVGO of £77.05, indicating that nearly 49.4% of the stock price is based on future growth expectations.

PVGO Sensitivity to EPS and Cost of Equity

Sensitivity Table

		Earnings Per Share£				
	77.05	2.84	3.84	4.84	5.84	6.84
Cost of Equity	8.00%	120.50	108.00	95.48	83.00	70.50
	7.00%	115.43	101.14	86.83	72.57	58.29
	6.13%	109.69	93.39	77.05	60.78	44.47
	5.00%	99.20	79.20	59.16	39.20	19.20
	4.00%	85.00	60.00	34.95	10.00	-15.00

This table highlights the linear impact of rising earnings on intrinsic value. For each £1 increase in EPS, the share price increases by approximately £15–£20, emphasizing the stock's strong sensitivity to earnings growth assumptions. It also shows that the no-growth valuation is highly sensitive to the cost of equity. A 2% increase in the discount rate (from 6.13% to 8%) decreases the no-growth value by over £18, significantly shrinking the PVGO component.

Comparable Company Analysis

The Comparable Company Analysis (CCA) is a widely utilized method within equity valuation, particularly useful for establishing a relative market value for a target firm based on the trading multiples of comparable businesses. In this analysis, the goal was to assess the relative valuation of Games Workshop Group plc (GAW) within the gaming and hobbyist retail sector. By comparing GAW to a selected group of peers over a five-year period (2020–2024), this section aims to illustrate how the market values GAW in the context of industry dynamics, earnings potential, and capital structure.

Peer Group

The companies selected for comparison were:

- **Frontier Developments plc**
- **Everplay Group plc**
- **Hornby plc**

The Rationale for selecting these peers has been discussed in Chapter 2 on page no. 10-12.

Methodology and Metric

For each company, financial and valuation metrics cover the years 2020 to 2024. These include:

- Enterprise Value (EV): Representing the market value of equity plus net debt.
- Revenue
- EBITDA
- Net Income
- Valuation Multiples:
    - EV/EBITDA

- EV/Revenue
- Price-to-Earnings (P/E) Ratio
- Price-to-Book (P/B) Ratio

These multiples are widely regarded as indicators of how investors price companies relative to performance. Multiples were benchmarked annually to reveal how valuation expectations have evolved.

Annual Financial and Valuation Overview

Year	Valuation Approach	Peer Median Multiple	GAW Metric	Implied Value
2020	EV/EBITDA	41.58	25.53	1061.56
2020	EV/REVENUE	10	9.38	93.80
2020	P/E	50.61	35.57	1800.20
2020	P/B	8.36	19.02	159.01
2021	EV/EBITDA	33.72	24.30	819.40
2021	EV/REVENUE	10.59	10.61	112.36
2021	P/E	46.09	31.90	1470.27
2021	P/B	8.65	19.87	171.88
2022	EV/EBITDA	17.15	14.07	241.30
2022	EV/REVENUE	4.13	5.76	23.79
2022	P/E	35.61	18.35	653.44
2022	P/B	4.01	10.05	40.30
2023	EV/EBITDA	5.67	16.37	92.82
2023	EV/REVENUE	1.64	6.52	10.69
2023	P/E	-7.15	23.08	-165.02
2023	P/B	1.19	13.24	15.76

2024	EV/EBITDA	-14.35	14.95	-214.53
2024	EV/REVENUE	1.47	6.22	9.14
2024	P/E	-5.34	21.56	-115.13
2024	P/B	1.34	7.63	10.22

Implied Valuation from Multiples

Using median peer multiples as benchmarks, the valuation sheet includes implied enterprise values for GAW based on each year's prevailing multiple. For example, in 2022, the peer median EV/EBITDA was 17.15 and GAW's EBITDA was 14.07, implying an enterprise value of approximately £241.3 million for that method. Similarly, valuation using P/E and EV/Revenue multiples resulted in a range of implied values, often well below GAW's actual market value—reflecting its premium relative valuation.

Interestingly, the 2023 and 2024 implied values began to decrease, driven more by declining sector medians than by GAW's operational performance. This highlights an important insight: peer-based valuation is sensitive to macro conditions and industry-wide investor sentiment, and in GAW's case, the company outperformed the sector even as peer multiples eroded.

Critical Assessment

The Comparable Company Analysis illustrates several key points:

1. Games Workshop Group consistently outperforms its peers in operational metrics, profitability, and investor confidence.
2. Valuation multiples for GAW remain strong but disciplined, suggesting market participants recognize the company's strengths without excessively overpricing its equity.

3. GAW's premium valuation is justified by its revenue growth, consistent dividends, strong margins, and resilient business model, especially during periods of economic uncertainty.

The Comparable Company Analysis underscores the relative strength and resilience of Games Workshop Group plc within its sector. While the broader industry exhibited shrinking multiples and declining enterprise values from 2022 to 2024, GAW maintained strong operating performance and premium valuation metrics. The company's consistent outperformance against peers in terms of revenue, EBITDA, and implied valuations validates its market position and justifies the higher multiples investors have been willing to pay. The data suggests that GAW's valuation is not only driven by sector metrics but is underpinned by unique business fundamentals, brand strength, and sustained profitability.

Nevertheless, several methodological limitations must be acknowledged. The peer group is relatively small and exhibits heterogeneous business models ranging from digital game development to toy and model manufacturing which introduces structural differences that may distort direct comparability. Additionally, negative or extreme multiples in certain years reduce the reliability of P/E based valuations during those periods. For this reason, EV/EBITDA and EV/Revenue multiples are considered the most robust indicators of relative value, as they are less sensitive to earnings volatility and capital structure differences. P/E and P/B should be treated as supplementary checks, applied with caution and contextualised within the broader financial and strategic environment.

In conclusion, the comparable company analysis suggests that Games Workshop commands a consistent valuation premium relative to its peers, driven by its distinctive intellectual property, robust profitability, and strong brand loyalty. However, the narrowing differential in recent years signals a shift in market expectations and underlines the importance of future growth execution. While EV/EBITDA and EV/Revenue provide the most reliable benchmarks, the significant dispersion in P/E and P/B outcomes highlights the limitations of relying solely on relative valuation in this context.

Accordingly, this analysis should be viewed as a complementary indicator of value rather than a definitive measure, best employed alongside intrinsic approaches such as discounted cash flow analysis to form a more comprehensive assessment of Games Workshop's fair value and strategic positioning within the sector.

Comparative Analysis and Discussion of Valuation Models

In order to ensure a robust and multidimensional assessment of Games Workshop's intrinsic value, this dissertation employed five core valuation methodologies: Free Cash Flow to the Firm (FCFF), Free Cash Flow to Equity (FCFE), Dividend Discount Model (DDM), Comparable Company Analysis (CCA), and Present Value of Growth Opportunities (PVGO). Each model captures different aspects of the company's financial architecture, investor expectations, and market position. This section integrates and compares these findings, offering critical insight into their convergence and divergence.

Overview of Valuation Outputs

The table below summarizes the key valuation results produced by the models applied:

Valuation Model	Implied Share Price	Upside vs Market Price	Key Assumption Basis
FCFF – DCF	£167.38 (16,738 pence)	~7.3%	Long-term free cash flows + terminal value
FCFE	Qualitative: Growing FCFE	Implied Upside	Net income and cash to equity holders
DDM	£196.61 (19,660 pence)	~26%	Future dividends and equity cost
Comparable Company Analysis	Market-aligned or Neutral	Minimal	Relative to sector peers
PVGO	£77.05 of £156 Price = Growth	Indicates high growth premium	Market-driven growth expectations

The most optimistic estimate stems from the Dividend Discount Model (DDM), while the most conservative is from the FCFF-based DCF model. The PVGO analysis supports the DDM by showing that approximately 49.4% of the share price is growth-driven, validating the use of future-oriented valuation frameworks.

Thematic Convergence: Undervaluation Thesis

All intrinsic valuations, despite variances in methodologies, reach the same conclusion; Games Workshop (GAW) shares are trading at undervalued prices versus modelled fair values.

The DCF model indicates intrinsic value of £167.38; the current equity value of £156.00 indicates that this model is not capturing the Company's future CF generating capacity on modest growth and terminal values.

- The DDM uses historical dividends the Company has paid out and has an intrinsic value of £196.61 which indicates a 26% upside and verifies the importance of dividends as a factor in valuation process to value shares.
- The FCFE Valuation doesn't produce a separate price but visually demonstrates a consistent growth in annual equity cash flow in particular to its capacity to finance dividends (capital returns) or share buy-backs.
- The PVGO evaluation shows that only £78.95 of the current share price is attributable to current earnings while £77.05 is to growth (assumed growth). This growth assumption lays out the investment thesis of GAW's notable licensing, global expansion and monetization of its IP.

Divergence in Magnitude: DDM Vs. DCF

The difference in magnitude between the DDM and DCF (FCFF) valuations is significant (£196.61 vs £167.38) and the difference can be largely considered conforming to the different purposes served by the respective models.

The DCF values are also indicative of risk by virtue of representing the reinvestment assumptions, working capital fluctuations, and operating (firm specific) risk which consequently means they exhibit sensitivity to macro variables and firm execution risk. It takes a holistic enterprise lens.

- The DDM, conversely, isolates cash returns to equity holders while assuming a stable payout regime. Historically, this assumption is true for Games Workshop given the firm's low leverage and conservative approach to reinvestment; conceivably, that dividends are a more direct representation of investor value.

However, the DDM is also highly sensitive to small variations in cost of equity or growth rate. A 0.5% change in either variable can create a valuation swing of 10--15%. Thus, while the implied upside is greater, the model risk is also higher.

Relative Valuation via Comparable Company Analysis

While the intrinsic models above rely primarily on income statement fundamentals, the Comparable Company Analysis (CCA) differs in reliance on multiples assigned by the market. The CCA results were relatively benign. Games Workshop was consistently able to trade at leverage premia relative to peers, however

- EV/EBITDA: Games Workshop ranged from 14-25x, indicating significant confidence from investors. However, peers in this set -- especially Hornby and Everplay -- exhibited distorted or negative multiples due to inconsistent earnings, raising questions about the reliability of relative valuation.
- P/E Ratios: GAW's P/E multiple of 21.56x (2024), while diminishing compared to its own historical peak, exceeds the respective medians of its peers, and affirms the purview of quality premia.

Importantly, relative valuation adds the dimension of market discipline, while also emphasizing that Games Workshop's elevated valuation is a result of its operational excellence and protective IP moat. For example, relative valuation does not account for the firm's unique dividend and growth profile, and is also influenced by cycles in the stock market and volatility in peers.

Strategic Insights from PVGO

This analysis also utilized Present Value of Growth Opportunities (PVGO) analysis. The PVGO model helps add clarity to the analysis by revealing the portion of the current share price that is based on future expected growth, particularly in design patents, licensing (Amazon Studios), future digital monetization, the expansion of international stores, and expectant demand cycles from shareholder engagement (e.g., Warhammer 40K universe). Overall, the PVGO helps corroborate the demonstrable high implied valuations of DDM and FCFE which are dependent on anticipated dividends and cash flows to equity. Of note, the PVGO adds inverse credence that valuations also have a forward-looking premium, and as such, represent all future opportunities with the accompanying risk of strategic mistakes or disintermediation by competitors.

Overall Assessment: Strengths and Applicability of Models

Model	Strengths	Limitations	Applicability to GAW
FCFF	Broad operational view; includes reinvestment risks	Sensitive to WACC and terminal assumptions	Appropriate for firm-level valuation
FCFE	Reflects actual equity-holder value	No single output; depends on debt policy assumptions	Good for assessing dividend capacity
DDM	Dividend-focused; simple and intuitive	Highly sensitive to inputs	Highly suitable given GAW's dividend history
CCA	Market-based; quick relative benchmarking	Distorted by peer underperformance or macro conditions	Best used as corroborative
PVGO	Captures investor sentiment and growth expectations	Backward-looking; relies on EPS and market price	Strategic overlay, not standalone

Summary and Recommendation

In summary, the valuation of Games Workshop Group across multiple models indicates a consistent undervaluation relative to its intrinsic worth. While each model carries inherent limitations, their collective convergence toward a higher fair value strengthens the reliability of the conclusion.

- The DCF and DDM provide the most coherent and data-grounded valuation estimates, both indicating material upside.
- The FCFE and PVGO models add strategic context, emphasizing the sustainability of shareholder returns and investor expectations for long-term growth.

- The CCA, though somewhat constrained by peer performance, still reinforces the market premium assigned to GAW's equity.

Thus, a triangulated valuation range of £167–£197 per share is supported by both cash flow models and market-based indicators. Given a current market price of £156, this suggests Games Workshop offers a compelling investment case — one grounded in strong fundamentals, scalable growth, and disciplined capital allocation.

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Note: The Financials of the target company and its peers were extracted from CAPITAL IQ.